

[H.P.G.C.L.]

Petition for
True-up of FY 2020-21,
Mid-Year Performance Review for FY 2021-22
&
Generation Tariff for FY 2022-23

Submitted to

Haryana Electricity Regulatory Commission

NOVEMBER 2021



Haryana Power Generation Corporation Limited

(An ISO: 9001, ISO: 14001 and OHSAS: 18001 Certified Company)

Corporate Identity Number: U45207HR1997SGC033517

Urja Bhawan, C-7, Sector-6, Panchkula, Haryana

Haryana Electricity Regulatory Commission

(Bay No 33-36, Sector-4, Panchkula, Haryana)

Case No:_____ of _____

IN THE MATTER OF

**Haryana Power Generation Corporation Limited
C-7, Urja Bhawan, Sector-6, Panchkula.**

....Petitioner

VS

Uttar Haryana Bijli Vitran Nigam Limited

Dakshin Haryana Bijli Vitran Nigam Limited

(Through Haryana Power Purchase Centre

(A Joint forum of DISCOMs))

Shakti Bhawan, Sector-6, Panchkula

....Respondents/ Beneficiaries

**Petition under Section 86 read with Section 61 & 62 of the
Electricity Act, 2003 for True-up of FY 2020-21, Mid-Year
Performance review for FY 2021-22 & seeking generation tariff
for FY 2022-23 as per MYT, Regulation 2019.**

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Notes and abbreviations

In this Petition:

- ✓ *Current Year is defined as Financial Year 2021-22*
- ✓ *Previous Year is defined as Financial Year 2020-21*
- ✓ *All currency figures used in this Petition, unless specifically stated otherwise, are in Rs. Cr.*

Abbreviation	Full Description
AAC	Additional Auxiliary Consumption (power consumed on account of RSD)
A&G	Administrative & General
APCPL	Aravali Power Corporation Private Limited
APC/AEC	Auxiliary Power/Energy Consumption
ARR	Aggregate Revenue Requirement
ATE	Appellate Tribunal for Electricity
CAGR	Cumulative Average Growth Rate
CERC	Central Electricity Regulatory Commission
CLP	China Light & Power (Jhazzar Power Limited)
Cr.	Crore
DCRTPP	Deen Bandhu Chotu Ram Thermal Power Plant
DHBVN	Dakshin Haryana Bijli Vitran Nigam Limited
DSI	Dry Sorbent Injection
DSM	Deviation Settlement Mechanism
EA – 2003	The Electricity Act 2003
FGD	Flue Gas Desulphurisation
FPA	Fuel Price Adjustment
FGPS	Faridabad Gas Power Station
FY	Financial Year
GCV	Gross Calorific Value
GFA	Gross Fixed Assets
GoH	Government of Haryana
Gol	Government of India
HERC	Haryana Electricity Regulatory Commission
HPGCL	Haryana Power Generation Corporation Limited
Ind AS	Indian Accounting Standard
IoB	Indian Overseas Bank
MoC	Ministry of Coal, Government of India
MoD	Merit Order Dispatch
MoEF&CC	Ministry of Environment, Forest and Climate Change
MoP	Ministry of Power, Government of India
MU	Million Units
MYT	Multi Year Tariff
O&M	Operation & Maintenance
PoC	Point of Connection
PFC	Power Finance Corporation
PLF	Plant Load Factor
PNB	Punjab National Bank
PPA	Power Purchase Agreement
PTPS	Panipat Thermal Power Station
REC	Rural Electrical Corporation
RGTPP	Rajiv Gandhi Thermal Power Plant
RSD	Reserve Shut Down
R&M	Repair & Maintenance
SBI	State Bank of India
SCR	Selective Catalytic Reduction
SFC	Secondary Fuel Consumption
SHR	Station Heat Rate
SLDC	State Load Dispatch Centre
SNCR	Selective Non Catalytic Reduction
SOFA	Separated Over Fire Air
SPM	Suspended Particulate Matter
STP	Sewage Treatment Plant
TO	Tariff Order
UHBVN	Uttar Haryana Bijli Vitran Nigam Limited
WYC	West Yamuna Canal

List of Events

Date	Description
31.10.2019	HERC notified HERC (Terms and Conditions for Determination of Tariff for Generation, Transmission, Wheeling and Distribution & Retail Supply under Multi Year Tariff Framework) Regulations, 2019 for the MYT Control Period of FY 2020-21 to FY 2024-25.
31.10.2019	As per Regulation 71 under HERC MYT Regulations, 2019, the ARR revenue requirement details and Tariff filing for determination of generation tariff for the ensuing year should be filed each year by 30 th November before Haryana Electricity Regulatory Commission.

BEFORE THE HARYANA ELECTRICITY REGULATORY COMMISSION, PANCHKULA

Filing No_____

Case No_____

IN THE MATTER OF:

**Filing of the Petition for the approval of True-up of
FY 2020-21, Mid-Year (Performance) Review for FY 2021-
22 & Generation Tariff for FY 2022-23 for HPGCL under
Section 61 and 62 of Electricity Act 2003**

..... Petitioner

The Petitioner respectfully submits as under: -

1. Haryana Power Generation Corporation Limited (HPGCL) is a Company incorporated under the provisions of the Companies Act, 1956 and is wholly owned by the Government of Haryana.
2. HERC notified HERC (Terms and Conditions for Determination of Tariff for Generation, Transmission, Wheeling and Distribution & Retail Supply under Multi Year Tariff Framework) Regulations, 2019 (hereinafter referred as HERC MYT Regulations, 2019) for the MYT Control Period of FY 2020-21 to FY 2024-25. As per Regulation 71 under HERC MYT Regulations, 2019, the revenue requirement details for determination of generation tariff for the ensuing year should be filed each year by 30th November before Haryana Electricity Regulatory Commission (hereinafter referred as "HERC" or "Hon'ble Commission").
3. In accordance with the provisions of Section 61 and 62 of Electricity Act, 2003 and as per the requirements and timeline given in Regulation 71 of HERC MYT Regulations, 2019, HPGCL hereby submits this petition for approval of the true-up for FY 2020-21, midyear performance review for FY 2021-22 and generation tariff determination for FY 2022-23.
4. Hon'ble Commission in its previous orders has not carried out mid-year review and restricted its order for the true up for previous year and tariff determination for the coming year. However, in compliance to the provision of Regulation 11 of the HERC MYT Regulation, 2019, HPGCL submits performance parameters for FY 2021-22 for reference of the Hon'ble Commission.
5. HPGCL has opted for appeals against various orders of the Hon'ble Commission and reserve its right to seek relief, if allowed by the Appellate authority in the matter. The details of appeals filed by HPGCL are tabulated as under:

S. No.	Name of the Appellate Authority	Ref. of the order against which appeal has been preferred	Matter/ Period to which appeal pertains	Appeal No./ Case Ref.
1.	Hon'ble APTEL	HERC Order in PRO-76 of 2020	FY 2019-20 (True Up) & 2021-22 (ARR)	APL 150/2021
2.	Hon'ble APTEL	HERC order dated 24.04.2020	FY 2018-19 & 2020-21	DFR 216 of 2020
3.	Hon'ble APTEL	HERC order dt. 31.03.2016	FY 2014-15 recovery of fixed cost	APL 306/2016
4.	Hon'ble Supreme Court of India	APTEL order dt. 18.09.2015	FY 2013-14 &ARR for control period 2014-17	CA 14027-28/2015
5.	Hon'ble Supreme Court of India	APTEL order dated 1.03.2012	FY 2010-11	CA 4831/2012
6.	Hon'ble Supreme Court of India	APTEL Order dated 31.07.2009	FY 2007-08	CA 2862/2010
7.	Hon'ble High Court of Punjab and Haryana	MYT Regulations 2012	Provisions of technical & commercial parameters	CWP 16027 of 2014

Date: **.11.2021**

Chief Engineer
Planning-cum-Regulatory
HPGCL, Panchkula
(Authorised Signatory)

A1 BRIEF PROFILE OF THE COMPANY

- 1.1 Haryana Power Generation Corporation Ltd. (HPGCL), hereinafter also referred to as the “Petitioner”, is a company registered under the Indian Companies Act, 1956 and is wholly owned subsidiary of Government of Haryana.
- 1.2 HPGCL generating stations were envisaged as the base load stations for the State of Haryana and are obligated to supply its entire power only to the Haryana State DISCOMs as per the State Government notification dated 11.04.2008.
- 1.3 HPGCL presently owns and operate the following power plants in the state as mentioned in Table 1 below:

Table 1: Plants Operated by HPGCL

Particulars	Installed Capacity	Date of Commissioning	Make/ OEM of Generating unit
Panipat Thermal Power Station, Panipat	Unit No-6: 210 MW Unit No-7: 250 MW Unit No-8: 250 MW	31/03/2001 28/09/2004 28/01/2005	BHEL
	Solar PV:10 MW	01/11/2016	Sterling Wilson
DCRTPP, Yamuna Nagar	Unit No-1: 300 MW Unit No-2: 300 MW	14/04/2008 24/06/2008	SEC (China)
RGTPP, Hisar	Unit No-1: 600 MW Unit No-2: 600 MW	24/08/2010 01/03/2011	SEC (China)
Western Yamuna Canal Hydro Project (Yamuna Nagar)	Power House A Unit No-1: 8 MW Unit No-2: 8 MW Power House B Unit No-1: 8 MW Unit No-2: 8 MW Power House C Unit No-1: 8 MW Unit No-2: 8 MW	29/05/1986 13/06/1986 15/05/1987 01/06/1987 27/03/1989 18/04/1989	Fuji Limited (Now acquired by Voith Hydro Limited)
	Power House D Unit No-1: 7.2 MW Unit No-2: 7.2 MW	16/04/2004 12/05/2004	BHEL
Total Capacity	2582.4 MW		

- 1.4 Haryana State is having 25% equity through HPGCL in the joint venture with Govt of Delhi and NTPC for 1500MW Indira Gandhi Super Thermal Power Project(IGSTPP) situated at Jhajjar, Haryana.
- 1.5 HPGCL is nodal agency entrusted by Government of Haryana for facilitating Nuclear Power Corporation of India Limited (NPCIL) in setting up 2800 MW (4X700 MW) Nuclear Power plant in Fatehabad district of Haryana, presently under construction.
- 1.6 Development of Kalyanpur Badalpara Coal Block
- Kalyanpur Badalpara Coal Block in Dumka Distt. (Jharkhand) was allotted to HPGCL on 24.02.2016. Initially HPGCL approached 3 no. Govt. PSUs to carry out its detailed

exploration and preparation of its Geological Report, but no firm agreed to undertake the job. Then HPGCL floated open tender for this work for 4- times, but no firm participated in the render. After persuasion with Ministry of Coal, CMPDI (PSU), Ranchi agreed to carry-out the work and floated tenders 3-times for finalizing a contracting agency but again no firm participated in the tender. Main reason for reluctance of firms to work in coal black area was adverse law & order problem over there. Accordingly, the proposal for terminating the allocated coal block was sent to Ministry of Coal on 19.10.2020 with the approval of Chief Minister of Haryana. Meanwhile, in view of the interest shown by a mining firm for development of coal block , CMPDI has been requested to intimate whether it is worth to give ine more trial to explore the coal block depending upon the present ground situation of the coal block area, vide letter dated 22.10.2021. Reply of CMPDI is awaited.

A2 PERFORMANCE OF HPGCL STATIONS OVER THE YEARS

2.1 The current installed capacity of HPGCL is 2582.4 MW. The following sections present the unit-wise historical performance of HPGCL plants.

Gross Generation and PLF

2.2 Tables below shows the Historical unit wise annual generation in MU and PLF:

Table 2: Historical Annual Gross Generation (in MU)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	324.00	0.00	51.93	0.00
PTPS-7	1308.75	884.46	619.48	433.17
PTPS-8	1569.40	1088.33	547.08	406.01
DCRTPP-1	1346.78	1574.14	1316.67	410.17
DCRTPP-2	1974.87	1166.89	1294.75	766.77
RGTPP-1	1622.71	768.95	1230.98	1045.68
RGTPP-2	2229.48	1547.17	405.92	0.00
WYC Hydel	237.68	300.03	242.91	120.17

Table 3: Historical Deemed PLF (%) (PAF)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	99.41	100.00	89.71	100.00
PTPS-7	98.79	94.41	89.75	98.67
PTPS-8	98.85	99.80	99.24	99.13
DCRTPP-1	69.51	100.00	100.00	54.53 ¹
DCRTPP-2	98.48	65.96	99.55	99.50
RGTPP-1	83.48	63.36	90.47	99.51
RGTPP-2	96.49	99.94	47.14	0.00 ²

¹DCRTPP-1 availability has been impacted due to capital overhauling and rectification works.

²RGTPP-2 is not available due to rotor delay from OEM.

Table 4: Historical PLF (%)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	17.61	0.00	2.82	0.00
PTPS-7	59.76	40.28	28.29	39.45
PTPS-8	71.66	49.56	24.98	36.98
DCRTPP-1	51.25	59.74	50.10	31.13
DCRTPP-2	75.15	44.28	49.27	58.19
RGTPP-1	30.87	14.59	23.42	39.68
RGTPP-2	42.42	29.36	7.72	0.00
WYC Hydel	43.48	54.74	44.44	43.85

2.3 The less scheduling of HPGCL units is primarily attributed to less demand placed by the beneficiary i.e. Haryana DISCOMs to HPGCL. The same is beyond control of HPGCL.

2.4 HPGCL plants are facing frequent backing downs/ less scheduling due to the improper procedure of merit order dispatch being adopted by DISCOMs, which is primarily based on marginal cost savings and doesn't reflect the true cost of power to consumers.

The issue is presently under challenge at Hon'ble APTEL by HPGCL to provide equitable field for State Generators don't have implication of PoC Charges and Losses.

- 2.5 The frequent backing downs/ Start stop operations of HPGCL generating stations not only increase metallurgical failures/ degradation of plants but also severally affects the performance parameters of generating units. Less demand from the DISCOMs is sole reason for Haryana State Based Generators for low PLF as shown under:

Table 5: Historical PLF of State Based Generators

Year	APCPL	CLP	FGPS	HPGCL PTPS (7&8)	DCRTPP	RGTPP
2017-18	60.03	64.97	22.99	47.15	65.60	44.53
2018-19	56.51	60.18	16.51	65.71	63.20	36.65
2019-20	28.96	50.52	14.92	44.92	52.01	21.97
2020-21	47.85	46.99	25.08	26.63	49.68	15.57*

* The lower PLF of RGTPP is due to outage of Unit-2 due to bending of rotor and also impacted by Covid Scenario.

- 2.6 Number of Start and Stop operations due to the instructions of the beneficiary during past years is given in the below table:

Table 6: Historical start stop operations

	PTPS 6	PTPS 7	PTPS 8	DCRTPP 1	DCRTPP2	RGTPP1	RGTPP2
FY 2017-18	8	12	14	4	5	11	8
FY 2018-19	12	15	8	8	6	9	10
FY 2019-20	-	12	10	5	5	6	8
FY 2020-21	2	11	11	8	12	6	4
FY 2021-22 (upto 09/ 21)*	-	6	3	1	3	4	-

*As per Regulation 34 of the HERC MYT 2019, maximum cap for Start/Stop operation is fixed at 07 Nos, and HPGCL is on the verge of crossing the said limit in FY 2021-22 for Unit-7 of PTPS. HPGCL reserves its right to claim for the excess expenditure on account of higher Start/Stop operation, if any, at appropriate time under True up.

- 2.7 The historical trend of the backing down of HPGCL generating stations is as under:

Table 7: Historical Backing down (%)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	81.80	100.00	86.89	100.00
PTPS-7	39.03	54.13	61.46	59.22
PTPS-8	27.19	50.24	74.25	62.16
DCRTPP-1	18.26	41.11	51.02	23.40
DCRTPP-2	23.33	21.68	50.28	41.31
RGTPP-1	52.61	48.77	67.05	59.83
RGTPP-2	54.08	70.59	39.42	0.00

2.8 From the above table, it reveals that HPGCL generating plants have faced excessive backing down in the past and is showing continuous rising trend , touching of alarming level in first half of FY 21-22. The same needs to be seen by the Hon'ble Commission and a fair guidance may be imparted to the DISCOMs to utilize the State resources in optimum way instead of adding new sources.

2.9 It is submitted that such significant backing down has adversely impacted HPGCL's operations/ performance in the following ways:

- I. While HPGCL generating units are placed under Reserve Shut Down (RSD) by DISCOMs, certain essential auxiliaries need to remain on bar for keeping Units available which require continuous drawl of electricity from the grid in absence of its own generation. This leads to increase in O&M charges and in case the same exceeds the normative charges, same needs to be compensated accordingly.
- II. SHR degradation has also been observed on account of running of Units at low loading on account of less demand which needs to be compensated as per MYT Regulation 2019.
- III. The condition of minimum off take of Coal under FSAs leads to coal stock pile up and its handling issues on account of low scheduling is a major challenge. Any loss on account of the same is beyond the control of HPGCL and needs to be pass through.
- IV. The reference is invited to HERC order dated 30.03.2021 in HERC/PRO - 77 of 2020 & HERC/PRO - 78 of 2020, wherein at page 142, it was directed as under:

".....Additionally, while resorting to bidding or calling for expression of interest for power procurement the Discoms must ensure that the power under PPAs already approved by the Commission materialises and also the intra-State generator i.e. HPGCL's power plants are scheduled at least up to the critical minimum threshold before considering backing down.."

The above directive of the Hon'ble Commission needs to be adhered by the DISCOMs in true spirit, so that optimum utilisation of HPGCL units is there in interest of the Consumers.
- V. Frequent backing down/ Start stop operations affects the operational life cycle of plants resulting in higher Repair & Maintenance expenses due to metallurgical degradation/ frequent failures. The same needs to be considered as pass through beyond the permissible limits, if required.

Specific Oil Consumption

2.10 The Table below shows the Historical specific oil consumption:

Table 8: Historical Unit wise Specific Oil Consumption (ml/kwh)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	1.77	-	5.17	-
PTPS-7	0.62	0.60	0.96	0.77
PTPS-8	0.36	0.43	0.92	0.53
DCRTPP-1	0.92	0.26	0.28	0.13
DCRTPP-2	0.25	0.55	0.37	0.20
RGTPP-1	0.85	1.22	0.65	0.36
RGTPP-2	0.46	0.84	1.70	0.00

2.11 From the Sp. Oil cons. table no.8 read with PLF table 4 along with start- stop table 6, it reveals that oil consumption of the generating plant mainly depends upon its scheduling/PLF / no. of start - stop operations. HPGCL reserve its right to seek compensation on account of oil at True-up, if required.

Auxiliary Consumption

2.12 The Historical unit wise Auxiliary Consumption when HPGCL units were on bar is as under:

Table 9: Historical Unit wise Auxiliary Consumption On-bar (in %)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	8.13	*	9.69	*
PTPS-7	8.10	8.49	8.72	9.58
PTPS-8	7.98	8.41	8.54	9.07
DCRTPP-1	7.81	8.02	8.04	8.39
DCRTPP-2	7.89	8.10	7.70	8.08
RGTPP-1	5.84	6.49	5.48	5.71
RGTPP-2	5.48	5.48	5.64	--

* remained under RSD on account of no demand from DISCOMs

It is worth to apprise the Commission that HPGCL Auxiliary consumptions (when on bar) remains within the allowable limits after considering the compensation in Auxiliary consumption as per Regulation no 34 of MYT,2019 for low loading.

Station Heat Rate

2.13 The table below shows the Historical Unit-wise Station Heat Rate (SHR):

Table 10: Historical Unit wise Station Heat Rate (in Kcal/kwh)

	2018-19	2019-20	2020-21	2021-22 (Upto Sep 21)
PTPS-6	2540	*	2537	*
PTPS-7	2473	2476	2476	2471
PTPS-8	2468	2471	2480	2478
DCRTPP-1	2327	2328	2341	2338
DCRTPP-2	2319	2333	2342	2340
RGTPP-1	2461	2476	2431	2400
RGTPP-2	2419	2442	2461	0

* remained under RSD on account of no demand from DISCOMs

- 2.14 HPGCL has implemented various standard O&M practices including regular monitoring and review by the expert groups & also at various levels of the management. Resultantly, it is able to meet with regulatory norms of SHR despite low scheduling except RGTPP.

A3 INITIATIVES TO IMPROVE EFFICIENCY

Efficiency Improvements measures taken by HPGCL:

3.1 The Haryana Power Generation Corporation Ltd. (HPGCL) is embarked on a mission to improve the efficiency of its thermal power stations. For the same, HPGCL has taken following initiatives:

- a) Based on the O&M practices being followed at NTPC Thermal Power Stations, an Operation Review Team (ORT) comprising of experts in various fields of power plant operation & maintenance from HPGCL and Ex-NTPC experts, has been constituted. The ORT is required to visit each power plant every month and analyse the performance parameters, heat rate, efficiency and safety at micro level. The target of ORT is to bring the operating parameters as close to the design parameters as possible.
- b) The Chief Engineers of power stations are taking Daily Plant Meetings with the officers, to closely monitor the performance parameters and for sorting out day to day problems quickly. Measures to improve efficiency of power stations and methodology of their implementation are also discussed in these meetings.
- c) An Apex Committee comprising of Managing Director, Directors, all Chief Engineers and Financial Advisors meets regularly to critically analyse the activities of the above groups and the performance of the power stations.
- d) The Boiler Knowledge Team (BKT) has been constituted to inspect the quality of work in boiler during overhauling of units to ensure improvement in performance after overhauling. The BKT also analyses the problems of boiler being faced in any plant of HPGCL.
- e) Turbine Knowledge Team (TKT) has been constituted to inspect the quality of work in the turbines during overhauling of the units. The TKT also analyses the problems of Turbine being faced in any plant of HPGCL.
- f) C&I Knowledge Team (CIKT) has been constituted to find solution to the problems relating to Controls & Instrumental system, being faced in the Power Stations of HPGCL.
- g) Fuel Knowledge Team (FKT) has been constituted, to recommend monthly rake movement of linked coal companies to be submitted by each plant, based on available coal stocks/scheduling of plant and demand supply position of coal market. FKT also deliberates and recommend about flexible utilization of coal in order to minimize cost per 1000Kcal of coal which subsequently helps in reducing the coal cost and generation cost.

- h) Ash Knowledge Team (AKT) has been constituted to maximize utilization of the dry ash & pond ash from HPGCL Power Stations.
- i) Electrical Knowledge Team (EKT) has been constituted to find solution to the problems relating to Electrical system, being faced in the Power Stations of HPGCL.

Coal Linkage status of HPGCL:

Table 11: Existing Coal linkages

Generating Plant	Coal Company	Existing ACQ (Lac MT)
RGTPP, Hisar	MCL	9.00
	NCL	15.00
	CCL	18.34
	ECL	4.00
	TOTAL	46.34
DCRTPP	CCL	28.00
PTPS, Panipat	CCL	20.60
	BCCL	11.60
	WCL	2.30
	TOTAL	34.50
HPGCL as whole		108.84

A4 NEW ENVIRONMENTAL NORMS

Background:

- 4.1 The Ministry of Environment, Forest and Climate Change (MoEF & CC), Government of India, notified the Environment (Protection) Amendment Rules, 2015 (Amendment Rules, 2015) on 07.12.2015 & dated 19.10.2020 amending/introducing the standards for emission of environmental pollutants to be followed by Thermal Power Plants.
- 4.2 The following are the amendments made to the existing allowable level of emission;

Table 12: Standard for Emission as per New Environment Law

Pollutants (mg/Nm3)	COD before 2003		COD after 2003	
	< 500 MW	> 500 MW	< 500 MW	> 500 MW
Particulate Matter (PM)	100		50	
SOx	600	200	600	200
NOx	600		450	
Hg	0.03			
Sp. Water Cons.(M³/MWh)	3.5			

- 4.3 The following actions are required to be taken for compliance of new norms: -
- SO_x**: - Flue Gas Desulphurisation (FGD) is required to be installed to control SO_x – levels.
 - NO_x**: - Following are the technologies for control of NO_x levels: -
 - Combustion modifications i.e. installation of low NO_x burners and Separated Over Fire Air (SOFA) Damper.
 - Installation of Selective Catalytic Reduction (SCR) / Selective Non- Catalytic Reduction (SNCR).
 - SPM**: - ESP rectification/up-gradation needs to be carried out to control SPM levels.
- 4.4 Hon'ble Commission has already granted in-principle approval in HERC/PRO-59 of 2018 vide order dated 07.03.2019 for implementation of New Environmental Norms.

Installation/Status of pollution control equipment:

- 4.5 The Commission may take note that all Thermal Power Plants of HPGCL are complying with the environmental norms related to SPM, Specific Water Consumption and Mercury.

a) Status of FGD for DCRTPP and RGTPP:

- 4.6 NIT's were floated for installation of Wet limestone based FGD at RGTPP, Hisar and DCRTPP, Yamunanagar on 24.04.2019 & 27.06.2019 respectively. Agenda notes for installation of FGD at RGTPP, Hisar and DCRTPP, Yamunanagar were put up before State Level HPPC meeting held on 10.02.2020 under the chairmanship of Hon'ble Chief Minister, Haryana for negotiations and award of work. HPPC deferred both the cases. Subsequently, Govt. of Haryana decided retendering and participation of only those companies which are registered in India.
- 4.7 Accordingly, fresh NITs on DCB mode were issued on 29.09.2020 with due date of bid submission as 29.10.2020. Part-I (Techno-commercial) of DCRTPP and RGTPP NITs were opened on 02.02.2021 and 10.02.2021 respectively. Part-II of DCRTPP and RGTPP were opened on 22.02.2021 and 04.03.2021 respectively.
- 4.8 In case of DCRTPP Yamunanagar, M/s SMS India Pvt. Ltd., Gurugram emerged as the L-1 bidder with quoted rates as Rs 551,44,45,620/- inclusive of taxes and in case of RGTPP Hisar, M/s PES Engineers Pvt. Ltd., Hyderabad is the L-1 bidder with quoted rates as Rs. 665,52,00,000/- inclusive of taxes.
- 4.9 The agenda notes for award of work for installation of FGD at DCRTPP and RGTPP were placed before State Level HPPC in the meeting held on 12.06.2021. After detailed deliberations, the Committee directed that **the Gol notification dated 31.03.2021** requires to be thoroughly examined. Accordingly, both the agendas were deferred.

b) Status of FGD for Unit-6,7&8 PTPS, Panipat:

- 4.10 NITs on DCB mode for installation of Dry Sorbent Injection (DSI) system in Unit-6 and Unit-7&8, PTPS, Panipat were issued on 29.09.2020 with due date of bid submission as 30.10.2020. Part-I & Part-II were opened on 03.02.2021 & 12.03.2021 respectively.
- 4.11 In case of Unit-7&8, PTPS, Panipat, M/s Melco India Pvt. Ltd., Faridabad is the L-1 bidder with quoted rates as Rs. 74,34,00,000/- inclusive of taxes and M/s KC Cottrell India Pvt. Ltd., Gurgaon is the L-2 bidder with quoted rates as Rs. 76,98,32,000/- inclusive of taxes. The difference between Prices of L-1 and L-2 bidder is 3.55% and is less than 5%.
- 4.12 Agenda note for award of work for installation of DSI system at Unit-7&8, PTPS was placed before State Level HPPC in the meeting held on 12.06.2021. After detailed deliberations, the Committee directed that the **Gol notification dated 31.03.2021** requires to be thoroughly examined. Accordingly, the agenda was deferred.

4.13 In case of Unit-6, M/s KC Cottrell India Pvt. Ltd., Gurgaon is the L-1 bidder with quoted rates as Rs. 40,27,33,999/- inclusive of taxes. However, the case for installation of DSI System at Unit-6, PTPS, Panipat has been deferred by the WTDs, HPGCL.

4.14 **MoEF&CC vide notification dated 31.03.2021** (referred in para a&b above) has amended the Environment (Protection) Rules, 1986 as follows:

“A task force shall be constituted by Central Pollution Control Board (CPCB) comprising of representative from Ministry of Environment and Forest and Climate Change, Ministry of Power, Central Electricity Authority (CEA) and CPCB to categorise thermal power plants in three categories as specified in the Table-I on the basis of their location to comply with the emission norms within the time limit as specified in column (4) of the Table- I, namely:

Table-I

Sl. No.	Category	Location/area	Timelines for compliance	
			Non retiring units	Retiring units
(1)	(2)	(3)	(4)	(5)
1	Category A	Within 10km radius of National Capital Region or cities having million plus population.	Upto 31 st December 2022	Upto 31 st December 2022
2	Category B	Within 10 km radius of Critically Polluted Areas or Non-attainment cities.	Upto 31 st December 2023	Upto 31 st December 2025
3	Category C	Other than those included in category A and B.	Upto 31 st December 2024	Upto 31 st December 2025

Provision of environmental compensation has also been provided in the ibid notification which is as under:

Non-Compliant operation beyond the Timeline	Environmental Compensation (Rs. per unit electricity generated)		
	Category A	Category B	Category C
0-180 days	0.10	0.07	0.05
181-365 days	0.15	0.10	0.075
366 days and beyond	0.20	0.15	0.10

4.15 CPCB, MoEF&CC vide letter dated 27.08.2021 has intimated that the Task Force in its 3rd meeting held on 26.08.2021 has categorized the **RGTPP, Hisar and DCRTPP, Yamunanagar Thermal Power Plants in Category “C”** and the Unit-6,7&8 of Panipat Thermal Power Plant in Category “A”.

4.16 Therefore, timeline for compliance of new environment norms for Category “C” (non retiring units) are upto 31.12.2024 and for Category “A” (non retiring units) are upto 31.12.2022.

c) **Status of NOx for HPGCL Power Plants:**

HPGCL had decided to get the work of NOx reduction done through respective OEMs of the Power Plants i.e. M/s BHEL for PTPS, & M/s SEC for RGTPP and DCRTPP. Accordingly, formal enquiry to OEMs for detailed techno-commercial offer for reduction in NOx emissions was issued and their offers were received. The technical scope for PTPS, Panipat was finalized with BHEL. WTDs of HPGCL in its meeting held on 02.02.2021 considered and decided to

recommend the case to Utility Level High Powered Purchase Committee (ULHPPC) for consideration, negotiation and decision in respect of award of work to BHEL.

- 4.17 Meanwhile, case was put up for administrative approval of Govt. of Haryana to install Low NOx burners at DCRTTP, Yamunanagar and RGTPP, Hisar from OEM i.e. SEC, China. However, **Govt. of Haryana has desired for a competitive bidding process rather than to award the contract to OEM at negotiated rates.** WTD's of HPGCL in its meeting held on 16.08.2021 accorded approval for issuance of NIT for Supply and installation of De-NOx Combustion Modification System at HPGCL Thermal Power Plants. Accordingly, e-NIT has been issued in August 2021 for all plants with due dated in October 2021, which stands extended to November 2021.

A5 OPERATIONAL CONSTRAINTS OF HPGCL

A 5.1 HPGCL Scheduling Regarding:

1. The Hon'ble Commission's reference is invited to Table 5, indicating continuous low PLF of HPGCL Units which attracts short lifting Coal penalties, which is a burden on the Consumers of the State.
2. Hon'ble Commission while calculating/ deciding the Average Power Purchase Cost (APPC) for DISCOM always provide cost of HPGCL Units at Normative level. The same can be seen from the HERC order dated 30.03.2021(Page 137 to 139) as under:

Table No 13 : HERC APPROVED POWER PURCHASE QUANTUM AND COST FOR FY 2021-22

SOURCE	Quantum	Energy Charges	Energy Cost	Fixed Cost	Total Cost	Average Cost
	MU	Rs/ kWh	Rs. Million	Rs. Million	Rs. Million	Rs / kWh
NTPC						
Singrauli STPS	1307	1.48	1934	971	2905	2.22
Rihand STPS I	409	1.54	631	402	1033	2.52
Rihand II TPS	359	1.54	552	300	852	2.37
Rihand III TPS	353	1.52	535	624	1159	3.28
Unchhahar TPS I	53	3.28	175	89	263	4.95
Unchhahar TPS II	111	3.31	369	175	544	4.88
Unchhahar TPS III	60	3.37	202	121	329	5.49
Unchhahar TPS IV	195	3.02	588	447	1035	5.30
Anta CCPP	19	2.75	52	135	186	9.87
Auraiya CCPP	19	2.77	52	202	254	13.53
Dadri CCPP	71	2.69	192	225	416	5.83
Faridabad CCPP	290	2.78	804	2508	3312	11.43
Farakka STPS	83	2.80	234	102	336	4.03
Kahalgaoon I STPS	109	2.29	249	193	442	4.07
Kahalgaoon II STPS	331	2.17	719	568	1287	3.89
Kol Dam HPS	138	2.57	356	943	1299	9.40
Tanda TPS Stage - 2 (Unit - I)	89	2.74	245	53	298	3.34
NHPC						
Salal I HPS	507	0.65	328	900	1228	2.42
Bairasiul HPS	127	0.80	101	297	398	3.13
Tanakpur HPS	25	1.73	44	61	105	4.15
Chamera I HPS	337	1.19	403	326	729	2.16
Chamera II HPS	49	1.01	50	52	102	2.06
Chamera-III HPS	80	2.07	165	212	378	4.72
Dhauliganga HPS	56	1.28	72	104	176	3.11
Dulhasti HPS	105	2.89	303	299	601	5.74
Uri I HPS	137	0.86	118	198	316	2.31
Uri-II HPS	69	2.24	154	238	392	5.69
Sewa II HPS	25	2.78	71	105	176	6.92
Parbati-III HPS	58	1.62	94	276	371	6.35
SJVNL (Nathpa Jhakri)						
SJVNL (Nathpa Jhakri) HPS	287	1.19	343	406	748	2.60
Rampur HPS	69	2.43	168	199	367	5.32
THDC						
Tehri (THDC) HPS	211	2.15	453	578	1031	4.89
Koteshwar HPS	48	2.43	117	134	251	5.20
Nuclear Power Corp.						

(NPC)						
NAPP (Narora)	195	3.16	617	0	617	3.16
RAPP	545	3.65	1988	0	1988	3.65
HPGCL (as per HERC Order)	14914	3.52	52565	14141	66705	4.47
Shared Project						
BBMB HPS	2578	0.00	0	1812	1812	0.70
DVC						
Mejia TPS	438	2.90	1268	968	2235	5.11
Koderma TPS	417	2.62	1092	1259	2351	5.64
Raghunathpur TPS	689	3.26	2246	1024	3270	4.75
UMPP						
CGPL Mundra UMPP TPS	2510	1.92	4815	2189	7005	2.79
Sasan UMPP TPS	3278	1.30	4253	553	4805	1.47
Others (Long Term Sources)						
PTC Tala, HPS	40	2.27	90	0	90	2.27
PTC GMR Kamalanga TPS	1842	1.29	2376	2548	4924	2.67
PTC Baglihar HPS J&K	272	3.91	1061	0	1061	3.91
PTC Lanco Amarkantak TPS	1818	1.96	3561	2651	6212	3.42
PTC Karchamwangtoo HPS 376 MW	1399	1.83	2554	2995	5548	3.97
PTC Karchamwangtoo HPS New						
IGSTPP, Jhajjar (Aravali) TPS	2406	3.50	8417	8565	16982	7.06
Pragati Power CCPP	156	2.52	394	1404	1798	11.53
Adani Power Ltd. Mundra TPS	8545	2.49	21295	8394	29689	3.47
Teesta III HPS	0	0.00	0	0	0	0.00
MGSTPS, CLP, Jhajjar TPS	3441	3.60	12388	9113	21501	6.25
RE Power						
Bhoruka HPS	29	3.49	102	0	102	3.49
P&R Gogripur HPS	10	4.26	41	0	41	4.26
Puri Oil Mill HPS	14	3.67	50	0	50	3.67
Shahbad Sugar Mill	81	4.04	326	0	326	4.04
Naraingarh Sugar Mill	140	5.13	719	0	719	5.13
Ch. Devi Lal Sugar Mill	9	4.04	36	0	36	4.04
Haryana Co. Sugar Mill	54	6.39	344	0	344	6.39
Hafed Sugar Mill	9	4.04	36	0	36	4.04
Meham Sugar Mill	0	0.00	0	0	0	0.00
Starwire (biomass)	66	8.85	587	0	587	8.85
Gemco (biomass)	54	8.85	474	0	474	8.85
AB Grain (Biomass Co-gen)	34	8.97	301	0	301	8.97
Sri Jyoti (biomass)	64	8.97	571	0	571	8.97
Mor Bio (biogas)	8	7.18	59	0	59	7.18
Solar Projects				0		
SDS Solar (JNNSM)	2	5.66	9	0	9	5.66
C&S Solar (JNNSM)	2	5.66	9	0	9	5.66
Chandraleela Solar (JNNSM)	1	5.71	8	0	8	5.71
Sukhbir Solal (JNNSM)	2	5.66	9	0	9	5.66
Zamil Solar (JNNSM)	2	5.66	9	0	9	5.66
Siwana Solar	8	6.01	50	0	50	6.01
H.R. Mineral Solar (JNNSM)	2	5.66	9	0	9	5.66

Tayal & Co Solar (JNN SM)	2	5.66	9	0	9	5.66
VKG Solar (JNN SM)	2	5.66	9	0	9	5.66
Utrecht Solar (JNN SM)	2	5.66	9	0	9	5.66
Subhash Solar (JNN SM)	2	5.66	9	0	9	5.66
Balarch Solar (JNN SM)	2	5.66	9	0	9	5.66
JBM Solar	33	5.68	189	0	189	5.68
HPGCL Solar	17	4.89	81	0	81	4.89
SECI Solar	133	5.50	730	0	730	5.50
Wind Power (SECI) 1000 MW	298	2.71	807	0	807	2.71
Paddy Stubble PP (New Projects)	314	5.38	1691	0	1691	5.38
Amplus Sun Solutions Pvt Ltd	83	2.99	248	0	248	2.99
PM KUSUM Solar	25	3.34	85	0	85	3.34
ISTS Solar	489	2.47	1209	0	1209	2.47
HPGCL Solar (New)	90	2.99	268	0	268	2.99
Solar Power (Compt Bidding)	232	2.99	695	0	695	2.99
ISTS Wind Projects P II & III	1235	2.52	3111	0	3111	2.52
ISTS Wind & Hybrid)	2902	2.70	7836	0	7836	2.70
Gati/Dans/Shiga/IA HPS	1287	3.97	5104	0	5104	3.97
Parbati II NHPC	71	1.60	113	0	113	1.60
JBM (WtE)	43	8.47	367	0	367	8.47
Sainson Paper Indust.	20	4.15	84	0	84	4.15
RSL Distilleries	17	6.77	117	0	117	6.77
Cleantech Power	9	7.24	62	0	62	7.24
Ecogreen Energy	82	8.47	692	0	692	8.47
Tanda TPS Stage - 2 (Unit - II) NTPC	72	2.75	198	0	198	2.75
TOTAL	59720	2.67	159336	70056	229399	3.84

- It can be observed from the above, that while deciding the APPC for the DISCOMs, the Hon'ble Commission always consider the HPGCL units as "Base Load" plants which are required to be requisitioned at normative level and the balance requirement needs to be requisitioned from other sources. The same has not been adhered by DISCOMs in true spirit is the cause of grave concern for HPGCL.
- It has already been adjudicated and identified by the Hon'ble Commission that PoC Charges and Losses are not levied on the HPGCL Units in its order dated 18.02.2021 in PRO-76 of 2020 and thus have less liability of fixed nature on these projects. Thus, all efforts shall be there by DISCOMs to schedule HPGCL Units at normative level, as allowed by the Hon'ble Commission.

5. The directives of the Hon'ble Commission while considering HPGCL at APPC is well balanced approach that HPGCL Units are to be scheduled at optimum level and the coal penalties in respect of HPGCL Units be avoided in the matter. The Hon'ble Commission view is practical and far sighted, to avoid any major impact on the Consumers of the State. The same has also been directed in last ARR order of DISCOMs in its order dated 30.03.2021 as under:

"Additionally, while resorting to bidding or calling for expression of interest for power procurement the Discoms must ensure that the power under PPAs already approved by the Commission materialises and also the intra-State generator i.e. HPGCL's power plants are scheduled at least up to the critical minimum threshold before considering backing down.

The DISCOMs are directed to perform cost benefit analysis including trade-off between purchase of REC and RE Power before considering proposal to procure RE Power."

6. The precedence of considering scheduling of HPGCL Units at normative is there in every HERC ARR order on the basis of base Load station to be operative at normative and the same has not been adhered by the DISCOMs , which leads to the degradation of HPGCL Units due to erratic or no scheduling in the matter.
7. The issue of RGTPP Unit-2 is the best example in the matter, where the OEM has raised the concern of rotor bending (which requires to be replaced) that HPGCL Unit was supplied as base load station, not as a peaking station and apprehended in various meetings that if this trend continued, then there may be more major failures which may financially burden the HPGCL units and erode the public money at large. The recommendation of OEM is annexed as **Annexure-P1**.
8. Hon'ble Commission must ensure that the quantum stands approved in ARR while deciding APPC for the DISCOMs need to be exhausted first from every approved source before making any deviation to the proposed quantum for scheduling.
9. It is worth to apprise that, in case, DISCOMs restrict to schedule the Quantum approved under APPC in true spirit and any deficit afterwards shall be requisitioned from any other source, HPGCL shall be used as 'Base Load' as the money has already been allocated in APPC and is being paid by the consumer of the State. Thus, all the Commissions directive in respect of HPGCL shall be adhered in true spirit & thus, no burden will be there on consumers.
10. The Hon'ble Commission may take the note of the above and impart necessary directions to DISCOMs to first adhere the full quantum allowed from each source allowed in the ARR while deciding the APPC and any shortfall shall be requisitioned from other sources after exhausting the quantum's allowed under each source while deciding APPC.

A5.2: Unit-6 Scheduling; regarding:

1. Hon'ble Commission vide its order dated 18.02.2021 has not determined the tariff of Unit-6, PTPS and HPGCL has opted for an appeal against the said issue. The APTEL vide its order dated 24.09.2021 in APL 150/2021 has stayed the said part of the order and also allowed to recover the fixed cost as prayed in the stay application. The relevant Stay order is placed at **Annexure-P2**.
2. HPGCL at this juncture is not challenging any of the issue adjudicated by the Commission on Stay application, the issue is being raised merely for calling the attention of the Commission about the step treatment being given by DISCOMs to HPGCL Units.
3. HPGCL has raised the concerns vide letter 27.08.2021 , the same may be perused at **Annexure-P3** that power is being procured from the IEX at abnormal high prices and boxing up of Unit-6, PTPS even at offered price of Rs 3.675/KWh cannot be justified and is against the interest of the Consumers of the State.
4. SCPP in its 57th meeting dated 30.03.2021 has decided to utilise the Unit-6, PTPS during peak season of July,2021 to September 2021, reproduced as under:

“The matter was deliberated in the meeting and SCPP approved the same. Further, it was also approved that if required; power from PTPS Unit-6 may be scheduled on MTP as a merchant power from July 2021 to September 2021 in view of non availability of khedar-2 as per MoD”

5. It is clear from above, that SCPP has identified that Unit-6, PTPS can be utilised in merchant sale in case requirement of power is there in the State, till the matter is not decided.
6. HPPC vide its letter dated 20.05.2021 has sought rates of Unit-6, PTPS for scheduling the same under merchant sale in compliance of the SCPP decision dated 30.03.2021. Accordingly, HPGCL vide letter dated 01.07.2021 has communicated the price of Rs 3.675/KWh for the duration of July to September 2021.
7. Further, the reference is invited to HERC order in HERC/PRO - 77 of 2020 & HERC/PRO - 78 of 2020 dated 30.03.2021, wherein at page 142, it was directed as under:

“.....Additionally, while resorting to bidding or calling for expression of interest for power procurement the Discoms must ensure that the power under PPAs already approved by the Commission materialises and also the intra-State generator i.e. HPGCL's power plants are scheduled at least up to the critical minimum threshold before considering backing down..”
8. It is clear directions from HERC that the State Generator has to be scheduled before resorting to short term bidding or calling any expression of interest, which has not been carried by HPPC in true spirit.

9. HERC in many orders has categorically directed DISCOMs to take prior approval while resorting to Short Term purchase of power above APPC price. Thus, the DISCOMs have tweaked the Commission permission of day ahead power to resort to costly purchase from market on day to day basis. The same require to be noted in the matter and corrective steps are required to be taken in interest of consumer of State at large.
10. It is worth to apprise that as per the knowledge of HPGCL, no regulatory order or directive bars the scheduling of Unit-6, PTPS by DISCOMs at offered price, when the power is being procured from the IEX at abnormal high rate of Rs 12/KWh.
11. It is a matter of fact that present Employee cost of HPGCL, even if, Unit-6 is decommissioned has to be borne by DISCOMs and thus there would be saving of R&M cost only.
12. The Short Term power purchase by DISCOMs @ Rs 12/KWh (approx) instead of Scheduling 190MW of Unit-6 (Ex-bus) @ Rs 3.675/Kwh has lead to additional burden of Rs 2.827 Crore during 16.08.2021 to 18.08.2021.
13. It is worth to apprise that R&M cost of Unit-6, PTPS, is approximately Rs 8.5 Crore, and if these type of incidences are noted and verified, the amount saved in the matter can be used to serve the R&M cost of Unit-6, PTPS and scheduling of the same in turn will reduce the burden of short lifting coal penalties, thus making HPGCL Unit-6,PTPS virtually a free of cost plant.
14. Further, HPGCL vide letter dated 01.07.2021 has communicated the price of Rs 3.675/KWh for merchant sale. The reasons for not taking standing approval from the Commission for having cheaper stand by source for peak season in present scenario where Adani and CGPL have backed out from contracted obligations is of great concern and needs to be noted.
15. The reference is further invited to the HPPC notification dated 11.04.2008 where in the SCPP is having the mandate (refer S.No 16 (a)) to take strategic decision with respect to procurement of short term power and long term power for the State of Haryana.
16. Therefore, the matter of non-scheduling of HPGCL Unit-6, PTPS @ Rs 3.675/KWh available inside the state, instead of resorting to costly merchant power procurement from outside the state by additionally paying transaction charges and grid losses in prevailing high price scenarios from IEX is of great concern. Thus, DISCOMs may be directed to ascertain the reasons for non-scheduling of Unit-6, PTPS and resorting to power purchase from IEX.
17. The Hon'ble Commission may take note of the above and direct the DISCOMs to exhaust all concluded contracts before proceeding for any short term purchase.

A5.3: Deviation Settlement Mechanism (DSM)

1. The Hon'ble Commission vide its order dated 18.02.2021 at page No 101 has directed as under:

" DSM

The issue raised regarding implementation of DSM has been considered. The Commission is of the view that the tariff determined by the Commission is Unit Wise and not power plant wise i.e. PTPS Units 6 to 8, are considered as separate Units. Hence, DSM ought to be applicable Units wise as such. The issue of ABT Meters can be sorted out mutually between the STU and HPGCL. Hence, the Commission is not inclined to relax the relevant Regulations as prayed for."

2. As per the above directive of the Commission, the matter was taken up with the State Transmission Utility to implement the DSM Unit wise.
3. The reference is invited to Clause 6.5 of the Haryana Grid code (HGC) Regulations, 2009, which states that the generating companies etc who intend to use open access must install ABT meters at inter-utility exchange points, capable of recording the parameters in accordance with the Central Electricity Regulations (CEA Regulations).
4. The relevant clauses of Haryana Grid Code (HGC) Regulations, 2009 and CEA Regulations are reproduced as under:

" HGC, 2009

6.5 Tariff Metering

6.5.1 *The generating companies, CGPs, the transmission licenses and the distribution licensees and EHV consumers who intend to use open access provisions would need to install the meters suitable for Availability Based Tariff (ABT) at inter-utility exchange points which would record the parameters in accordance with the Central Electricity Authority Regulations on installation and operation of meters read with its amendment, if any.*

CEA Regulations, 2019

7. Location of meters-

(1) The location of interface meters, consumer meters and energy accounting and audit meters shall be as per the table given below:

Provided that the generating companies or licensees may install meters at additional locations in their systems depending upon the requirement.

<i>SR. NO.</i>	<i>STAGES</i>	<i>MAIN METER</i>	<i>CHECK METER</i>	<i>STANDBY METER</i>
1.	Generating Station	On all outgoing feeders including bus sectionalizer or tie line between two stages of generating stations having different tariffs or different ownership or both	On all outgoing feeders including bus sectionalizer or tie line between two stages of generating stations having different tariffs or different ownership or both	(i) High Voltage (HV) side of Generator Transformers (ii) High Voltage side of all Station Auxiliary Transformers.

5. From the perusal of the above relevant Regulations, it is very clear that ABT meters are to be installed on all outgoing feeders.

6. Term 'feeder' has been defined in clause 2(24) of the Haryana Electricity Regulatory Commission (Duty to supply electricity on request, Power to recover expenditure incurred in providing supply and power to require security) Regulations, 2005 as ;

"2. Definitions and Interpretations

.....

(24) "feeder" means an electrical line emanating from a substation to which a distribution substation or LT or HT consumers are connected;"

7. Thus, a plain reading of above makes it amply clear that a feeder is electrical line to which substations and consumers are connected, Further , interpreting the word 'outgoing', in its literal sense would mean that an 'outgoing feeder' is a feeder located at the outer periphery of premises of an entity.
8. The Hon'ble Appellate Tribunal for Electricity ("Hon'ble APTEL") in an Appeal filed by a State Transmission Unit seeking Notional loss from captive Generating Plant for grant of connectivity (Appeal No 209 of 2014 Judgement dated 29.05.2015), the operative portion of the judgment is reproduced here;

*"15. As per Regulation 7 of the CEA (Metering) Regulations, **the meter was to be placed at the outgoing feeder....**"*

9. Thus, it is clear from the above, that the meter has to be installed at the outgoing feeder of the HPGCL Units, which therefore can be only on plant basis and not on unit basis.
10. Further, HVPNL vide its letter dated 18.10.2021 along with a report (**Annexure-P4**), communicated to HPGCL the practical difficulties in installing ABT meters Units wise and detailed its reasons in the said report for installing the same on Unit wise:

"Feasibility Constraints:

1. In order to ascertain the **net energy units** generated by each generation unit, we need to subtract the energy units consumed by the auxiliary Station Transformers i.e., ST-1 and ST-2.
2. From the single line diagram, we can see that it is not possible to assign the energy units consumed by ST-1 and ST-2 to a particular generating unit, because there are some auxiliaries which are fed from both the sources i.e., ST-1 and ST-2 and common to both the units.
3. Also, from metering point of view the Generating units and Station Transformers doesn't have a dedicated PT but rather uses Bus PT via PT Selection relay which is not recommendable.
4. The meter at GT-1, GT-2, ST-1 and ST-2 can be replaced with ABT meters so that reading can be recorded every 15 minutes but in the existing infrastructure, it is not possible to have a dedicated PT for each generating unit and station Transformer.
5. Both the buses i.e., Bus-I and Bus-II are synchronized to maintain the reliability of the system both buses cannot be segregated.

Conclusion

As per the existing infrastructure/ arrangement at HPGCL generating stations, it is not feasible to achieve unit wise DSM, which requires unit wise consumption every 15 minutes.”

“Feasibility Constraints of DSM

1. In order to ascertain the actual injection on the bus by each generator i.e., Ex-bus generation of each unit, we need to subtract the energy consumed by the auxiliary Station Transformers i.e., ST-1 and ST-2.
2. At present condition of the plant, it is not possible to assign the energy units consumed by ST-1 and ST-2 from particular generating unit because there are some auxiliaries which are fed from both the Station Transformers i.e., ST-1 and ST-2 and common to both units.
3. The meters at GT-1, GT-2, ST-1 and ST-2 can be replaced with ABT meters so that readings can be recorded every 15 minutes but in the existing infrastructure, it is not possible to have a dedicated PT for each generating unit and station Transformers.
4. It is not feasible to feed the particular transmission line from a dedicated generating unit without loss of synchronism with grid.
5. Moreover, ST-1 & ST-2 are connected on the same bus where generators inject the power.

Conclusion

As per the existing infrastructure/arrangement at HPGCL generating stations, it is not feasible to segregate dedicated feeders to individual generating unit.””

11. The Commission also has approved various hydro sources which are situated outside the State and the tariff and other issues are governed under HERC Regulation, where the DSM/Plant availability is considered on plant basis only and the tariff of the same is governed on plant basis not on Unit basis. Thus, the equitable should be extended to HPGCL as per the Regulations of CEA/HGC.
12. In view of above, it is once again requested that as the plant meters are required to be outside the plant as per settled law, the plant base DSM and plant based tariff recovery be allowed to HPGCL in relaxation by exercising the Regulation 78 & 79 of the MYT Regulation, 2019.

A 5.4: Relaxation for Auxiliary Consumption in PTPS due to Tube Mill

- The reference is invited to HERC MYT Regulations where the norms for HPGCL Units for auxiliary consumptions are provided as under:-

“28. NORMS OF OPERATION FOR THERMAL POWER STATIONS

Normative Annual Plant Availability Factor (NAPAF)

A) Existing Plants

Plant name	2020-21	2021-22	2022-23	2023-24	2024-25
Panipat TPS (Unit-5)	85	-	-	-	-
Panipat TPS (Unit6)	85	85	85	85	85
Panipat TPS (Unit-7)	85	85	85	85	85
Panipat TPS (Unit8)	85	85	85	85	85
DCR TPS, Yamuna Nagar(Unit-1)	85	85	85	85	85
DCR TPS, Yamuna Nagar(Unit-2)	85	85	85	85	85
Rajiv Gandhi TPS, Khedar (Hisar	85	85	85	85	85
Rajiv Gandhi TPS, Khedar (Hisar	85	85	85	85	85

(B) New Plants Commissioned on or After 1st April 2020

Description	
Normative Plant Availability Factor (NAPAF)	85%

(2) AUXILIARY ENERGY CONSUMPTION

(a) Existing Plants

	MYT period				
Plant name	2020-21	2021-22	2022-23	2023-24	2024-25
Panipat TPS (Unit-5&6)	9	9	9	9	9
Panipat TPS (Unit7&8)	8.50	8.50	8.50	8.50	8.50
DCR TPS, Yamuna Nagar(Unit-1&2)	8.50	8.50	8.50	8.50	8.50
Rajiv Gandhi TPS, Khedar (Hisar) Unit-1&2)	6	6	6	6	6

(b) New Plants Commissioned on or after 1st April 2020

Description	Auxiliary Energy Consumption (%)
(i) Coal-based Generating Plants(with natural draft cooling tower or without cooling tower)*	
With Steam driven boiler feed pumps	6.00
With Electrically driven boiler feed pumps	8.50
(ii) Gas Turbine Generating Plants	
Combined Cycle	2.75
Open Cycle	1.00

For Coal-based generating stations with induced draft cooling towers, the norms shall be further increased by 0.5%.”

2. The Hon’ble Commission’s reference is invited to the issue of auxiliary consumption of PTPS units where the Tube mill is used.
3. The CERC in its tariff regulations 2019, while providing the “Norms of Operation” under Chapter 12 at “Auxiliary Energy Consumption” has allowed as under:

“Provided that for thermal generating stations with induced draft cooling towers and where the tube mill is used, the norms shall be further increased by 0.5% and 0.8% respectively”
4. The matter is of great concern for HPGCL as the allowed auxiliary consumption is creating constraint on PTPS Unit operations and may result in loss to HPGCL.
5. The State of Maharashtra and Rajasthan State Electricity Regulatory Commission (**Annexure P5**) has also allowed the auxiliary consumption in line with CERC regulations for tube mills. It is therefore requested that similar relaxation in norms may be provided by Commission by exercising its power under Regulation 78 & 79 of MYT Regulations 2019 and allow 0.8% of additional auxiliary consumption for PTPS units.
6. Hon’ble Commission is therefore requested to increase the Auxiliary Consumption of PTPS Units by 0.8%.

A5.5: Permission to increase stock of fuel from 30 days to 60 days for peak season:

1. The Hon'ble Commission must be aware that the coal shortage issue being faced by all the Generating Companies in present year created havoc for the power sector and forced to purchase costly power from the open market by DISCOMs due to non-availability of coal and HPGCL is not an exception to it.
2. The said shortfall created as per the articles prima facie due to non intimation of the DISCOMs to the generator their firm requirements and resorting to open market purchase on pretext of cheaper power and call for penalty of coal at later stage in the matter to attract more losses than savings.
3. HPGCL, to protect the consumer interest in the matter has written several communications to the DISCOMs that onset of monsoon season may lead to flooding of coal mines which in turn may reduce coal availability. Thus firm quantity to be scheduled from HPGCL was asked, so that necessary arrangement may be made for coal to support DISCOMs to serve the Consumers of the State. The same remains unanswered in the matter. The communication in the matter may be perused at **Annexure-P6**.
4. In view of the above, where HPGCL has also faced short fall in coal and situation become Touch and go for plants, it is need of the hour to allow HPGCL to increase the coal stock for peak season in the interest of Consumers of the State.
5. Therefore, relaxation is being sought in the matter to allow HPGCL to stock 60 days of Coal for meeting the demand of peak season by relaxing the norms under Regulation 78 & 79 of MYT Regulations.
6. It is further clarified that HPGCL is not seeking any increase in Interest on Working capital and be adhered with the approved norms; however the same if required be consolidated at the time of true-up. Thus, no burden on consumers is there in the matter. However, if the Commission feels that the stock of coal has to be as per the MYT Regulation, 2019, then the deemed availability charges on account of failure in declaration of availability on account of coal due to non confirmation of arrangement of fuel in advance by HPPC needs to be allowed in the matter, please.
7. The Hon'ble commission is requested to allow the same by exercising the regulation 78 & 79 of MYT Regulation, 2019.

A5.6: MOEFCC Notification dated 25.01.2016 under “Change in law” for Transportation of FLY Ash; regarding

1. The MOEFCC vide its notification dated 25.01.2016 has directed as under:

“ 2. In the said notification, in paragraph 2:-

(a) after sub-paragraph (1), the following proviso shall be inserted, namely:- “provided further that the restriction to provide 20 % of dry ESP fly ash free of cost shall not apply to those thermal power plants which are able to utilise 100 % fly ash in the prescribed manner.”

(b) after sub-paragraph (7), the following sub-paragraphs shall be inserted, namely:-

(8) Every coal or lignite based thermal power plants (including captive and or co-generating stations) shall, within three months from the date of notification, upload on their website the details of stock of each type of ash available with them and thereafter shall update the stock position at least once a Month.

(9) Every coal or lignite based thermal power plants shall install dedicated dry ash silos having separate access roads so as to ease the delivery of fly ash.

(10) The cost of transportation of ash for road construction projects or for manufacturing of ash based products or use as soil conditioner in agriculture activity within a radius of hundred kilometers from a coal or lignite based thermal power plant shall be borne by such coal or lignite based thermal power plant and the cost of transportation beyond the radius of hundred kilometers and up to three hundred kilometers shall be shared equally between the user and the coal or lignite based thermal power plant.

(11) The coal or lignite based thermal power plants shall promote, adopt and set up (financial and other associated infrastructure) the ash based product manufacturing facilities within their premises or in the vicinity of their premises so as to reduce the transportation of ash.

(12) The coal or lignite based thermal power plants in the vicinity of the cities shall promote, support and assist in setting up of ash based product manufacturing units so as to meet the requirements of bricks and other building construction materials and also to reduce the transportation.

(13) To ensure that the contractor of road construction utilizes the ash in the road, the Authority concerned for road construction shall link the payment of contractor with the certification of ash supply from the thermal power plants.

(14) The coal or lignite based thermal power plants shall within a radius of three hundred kilometers bear the entire cost of transportation of ash to the site of road construction projects under Pradhan Mantri Gramin Sadak Yojna and asset creation programmes of the Government involving construction of buildings, road, dams and embankments”

2. The above MOEFCC notification is “Change in Law” as per Regulation 3.20 of MYT Regulation 2019 and HPGCL is required to be compensated on account of expenditure incurred by HPGCL for transportation of the FLY ash.

3. The matter was adjudicated by CERC in the matter of NTPC vs others (in Petition No 172/MP/2016), wherein HPPC itself is the party in the matter. CERC has adjudicated that MOEFCC notification dated 25.01.2016 as a “Change in Law” event. The relevant para of the order is reproduced as under:

“31. Accordingly, we in exercise of the regulatory power hold that the actual additional expenditure incurred by the Petitioner towards transportation of ash in terms of the MOEFCC Notification is admissible under “Change in Law” as additional O&M expenses. However, the admissibility of the claims is subject to prudence check of the following conditions on case to case basis for each station:

- a) Award of fly ash transportation contract through a transparent competitive bidding procedure. Alternatively, the schedule rates of the respective State Governments, as applicable for transportation of fly ash.
- b) Details of the actual additional expenditure incurred on Ash transportation after 25.1.2016, duly certified by auditors.
- c) Details of the Revenue generated from sale of fly ash/ fly ash products and the expenditure incurred towards Ash utilization up to 25.1.2016 and from 25.1.2016 to till date, separately.
- d) Revenue generated from fly Ash sales maintained in a separate account as per the MoEF notification.

32. *The Petitioner is granted liberty to approach the Commission at the time of revision of tariff of the generating stations based on trueing –up exercise for the period 2014-19 in terms of Regulation 8 of the 2014 Tariff Regulations along with all details / information, duly certified by auditor”*

4. From the above CERC order, it is evident that the principle laid in the matter has been accepted by HPPC and becomes the settled principle that how the claim of fly ash transportation needs to be passed through under “Change in Law” by CERC.
5. HPGCL is also seeking relief on similar lines as already adjudicated by CERC and accepted by HPPC at par with NTPC in the matter. Thus, any amount incurred on account of transportation of ash needs to be compensated in the matter.
6. The Ministry of Power has also issued following instruction on dated 22.09.2021 to increase ash utilization:

“It is observed that the demand of Fly Ash has been increasing year on year basis contributing to increase in the Fly Ash utilization. Fly Ash is emerging as a valuable commodity. As the end users of Fly Ash like cement plants, brick kilns, road and construction agencies etc. are commercial ventures and all their input costs are accounted for, Fly Ash should invariable be auctioned through a transparent bidding process.

2. *All Coal/lignite based power plants are hereby advised to provide fly ash to the end users for all new commitments for supply of fly ash based on the following guidelines:*

2.1 *The power plants shall provide the Fly Ash to end users through a transparent bidding process only.*

2.2 *If after bidding/auction some quantity of Fly Ash still remains un-utilized, then only, as one of the options, it could be considered to be given free of cost on first come first served basis if the user agency is willing to bear transportation cost.*

2.3 *If ash remains unutilised even after the steps taken in Paras 2.1 and 2.2 above TPP shall bear the cost of transportation of Fly Ash to be provided free to eligible projects.*

2.4 *The end users shall be obligated to source the fly ash from the nearest TPPs to reduce the cost of fly ash transportation. If the nearest TPP refuses to do so, the end user project shall approach Ministry of Power for appropriate directions.*

2.5 *The transportation cost wherever required to be borne as per provisions of MOEF&CC notification by the power plants, shall be discovered on competitive bidding basis only. Thermal Power Plants shall prepare a panel of transportation agencies every year based on competitive bidding for transportation in slabs of 50km which may be used for the period. The TPPs shall call for bids well in advance so, that a transportation panel is in place as soon as the previous panel expires. There should not be gap between the expiry of one panel and the finalization of the fresh panel.*

2.6 *The fly ash will be offered to the end users on the competing demand basis, i.e. the end users who offer the highest price for fly ash and seek minimum*

support for transportation cost will be offered the same fly ash on priority. This will reduce the tariff of electricity and burden on the consumers.

2.7 The power plants may offer fly ash subject to their technical restrictions such as all precautions required for Dyke Stability and Safety etc. The power plants having lower ash utilizations shall make all out efforts to increase the fly ash utilization.

3. All concerned are requested to take necessary action in this regard.

4. This issue with the approval of the Hon'ble Minister of Power and NRE."

7. In view of the above, HPGCL is making all out efforts to dispose the ash free of cost, but in case, any amount is required to be incurred for transportation of ash, that needs to be compensated in the matter at the time of true-up.
8. The Commission is hereby requested to kindly consider the MoEFCC notification under "Change in Law" and allow HPGCL to be compensated in the matter as already accepted by HPPC with CGS i.e. NTPC under "True-up" on yearly basis.

A6 CURRENT TARIFF PETITION: PREMISE AND GUIDING PRINCIPLES

- 6.1 The Govt. of India notified the Electricity Act, 2003 (hereinafter referred to as the Act) with effect from 10th June 2003 repealing the Indian Electricity Act-1910, the Electricity (Supply) Act, 1948 and the Electricity Regulatory Commissions Act, 1998. The guiding principles for the power sector have been formulated by the Govt. of India in National Electricity Policy, 2005 and National Tariff Policy, 2016 under the power granted to it under Section 3 of the Electricity Act 2003. As per the National Tariff Policy, the Central Commission would, in consultation with the Central Electricity Authority, notify operating norms from time to time for Generation and Transmission which would be adopted by the SERCs. Under section 61 of the Act, the State Electricity Regulatory Commissions (SERC) are to be guided by principles and methodologies of Central Commission, National Electricity Policy, National Tariff Policy etc. while specifying the terms and conditions of determination of tariff.
- 6.2 The Hon'ble Commission has notified the HERC MYT Regulations 2019 for the control period 2020-21 to 2024-25. Accordingly, HPGCL has submitted the tariff for True up for FY 2020-21 and Mid Year Review of FY 2021-22 & Generation tariff determination for FY 2022-23 as per HERC MYT Regulation, 2019.
- 6.3 Hon'ble Commission didn't carry out Mid-Year Performance Review in its previous tariff orders; accordingly, HPGCL has not submitted the tariff impact for the midyear performance review for FY 2021-22. However, performance indicator has been submitted for the year for kind consideration of the Hon'ble Commission.

A7 CAPITAL INVESTMENT PLAN

7.1 Hon'ble Commission vide its Order dated 18.02.2021 has disposed off HPGCL petition PRO 76 of 2020 with the directives regarding financial progress of Capex approved by the Commission including any work wise deviation from the same. Accordingly, the Approved Capital Investment Plan of HPGCL was as under:

Table 14: Approved Capital Investment Plan

S. No.	Capital Expenditure Work	(Rs. Cr.)				
	Year	2020-21	2021-22	2022-23	2023-24	2024-25
1	Capital Overhauling at WYC	3.5	-	-	-	-
2	ERP System and allied works	-	31.26	-	-	-
3	Data Centre, Data Recovery centre etc. for ERP Solution	-	13.38	-	-	-
4	Balance Payment to R-Infra against EPC contract for RGTPP, Hisar	-	9.43	-	-	-
5	Procurement of PA fan blades for RGTPP Hisar	0.42	-	-	-	-
6	Procurement of 2 No. Air Driers for Transport Compressors for RGTPP Hisar	-	-	0.75	-	-
7	Construction of 2 no. Barracks for CISF for RGTPP Hisar	1.28	-	-	-	-
8	Installation of CCTV surveillance System in RGTPP Hisar	-	2.78	-	-	-
9	Construction of DAV school in power plant colony for RGTPS Hisar	-	-	6.87	-	-
10	Revival of 02 Nos of ESP fields of RGTPP Unit I	6.17	-	-	-	-
11	Improvement work of Cooling Towers of RGTPP Unit I & II	6.00	-	-	-	-
12	Installation of Variable Frequency Drive in Condensate Extraction Pump (CEP) of RGTPP Unit I & II	-	5.21	-	-	-
13	Replacement of 2 Nos. Stators of BCP of RGTPP Unit I & II	-	3.16	-	-	-
14	Up gradation of C&I system for RGTPP Hisar	3.00	-	8.0	-	-
15	Procurement of ID fan blades, RGTPP	-	1.68	-	-	-
16	Replacement of 03 Nos. Fire Tenders at RGTPP	-	-	1.20	-	-
17	Up gradation of hardware and software of PLC at RGTPP, Khedar, Hisar	-	-	-	-	7.00
18	Replacement of 2 Nos. (one for each unit) Battery Banks for main plant 2x150 kVA UPS System for Unit 1 & 2, RGTPP, Khedar, Hisar	0.60	-	-	-	-

S. No.	Capital Expenditure Work	(Rs. Cr.)				
		2020-21	2021-22	2022-23	2023-24	2024-25
19	Procurement of Complete Battery Banks Lead Acid Plante 220V, 2140AH in each Unit (Unit 1&2), RGTPP, Khedar, Hisar	1.90	1.9	-	-	-
20	Work for Supply, Erection, Testing and Commissioning of 02 Nos. ABB make unitrol-6080 Digital Automatic Voltage Regulator (DAVR) for Generator Excitation System and replacement with existing ABB make Unitrol-F DAVR at RGTPP, Khedar, Hisar	-	-	-	1.01	-
21	Construction of First Aid Centre and additional RCC Roof slab of DG Set house at RGTPP, Khedar, Hisar	-	0.55	-	-	-
22	Purchase of Ion Chromatography system fully automatic PC based, RGTPP Hisar	-	0.65	-	-	-
23	Revival of Fire Fighting System of Unit6, PTPS, Panipat	-	-	0.6	-	-
24	Replacement of damaged floor and Construction of Roads in PTPS Colony, Panipat as per new norms of Government of Haryana	-	1.55	-	-	-
25	Up-gradation of PTPS Unit-6 HMI System of pro-control supplied by M/s BHEL	-	-	1.5	-	-
26	Energy Management System PTPS Unit- 7-8	-	-	0.7	-	-
27	Modernization of Boiler Lift for PTPS Unit 8	-	-	0.55	-	-
28	Replacement of 02 Nos. Fire Tenders at PTPS Panipat	0.4	0.4	-	-	-
29	Renovation of centralised AC System of Unit-7&8,PTPS Panipat	-	-	1.8	-	-
30	Providing rejected Coal (Pucca Floor under 132 KV & 220 KV Lines inside the plant boundary) PTPS, Panipat	0.5	-	-	-	-
31	Construction of all weather patrolling track along the peripheral boundary wall at PTPS, Panipat	0.5	-	-	-	-
32	Replacement of 8" water lines around the circular road in PTPS, Colony	0.5	-	-	-	-
33	Replacement of PTPS Unit-6 AD Line in Ash Handling & repair D2 of ESP Field	0.50	-	-	-	-

S. No.	Capital Expenditure Work	(Rs. Cr.)				
	Year	2020-21	2021-22	2022-23	2023-24	2024-25
34	Up gradation of existing DCS system for DCRTTP 1 & 2	4	-	-	-	-
35	Township for DCRTTP, Yamunanagar	-	2.36	-	-	-
36	Providing of 2 No. VFD on Unit-II DCRTTP ,6.6KV Motor of CEP	-	2.36	-	-	-
37	Up-gradation of existing PLC & SCADA at DCRTTP	-	2.25	-	-	-
38	Refurbishment of BFP Cartridge: DCRTTP, Yamuna Nagar	-	1.6	-	-	-
39	Purchase of 01 no. Runner Hub without blades and new set of guide vanes	-	-	7	-	-
	Total	29.27	80.52	28.97	1.01	7.00
	Grand Total FY 2020-21 to FY 2024-25	146.77				

7.2 There are certain variations in the actual CAPEX incurred vis-à-vis approved expenditure tabulated above in Table14 mainly due to revision in the overhauling schedule/ financial prudence and some of the schemes have been completed/surrendered in the FY 2020-21.

7.3 In view of the above, the revised schedule of capitalization of the rest of the capital works is presented below for kind consideration and approval of the Hon'ble Commission.

Table 15: Revised capital investment plan

Sr No	Capital Expenditure Work	(Rs. Cr.)			
	Year	2021-22	2022-23	2023-24	2024-25
1	Capital Overhauling at WYC	3.50			
2	ERP System and allied works		31.26		
3	Data Centre, Data Recovery centre etc. for ERP Solution		13.67		
4	Balance Payment to R-Infra against EPC contract for RGTPP, Hisar			4.43	5.0
5	Construction of 2 no. Barracks for CISF for RGTPP Hisar	0.72			
6	Installation of CCTV surveillance System in RGTPP Hisar	2.44			
7	Construction of DAV school in power plant colony for RGTPS Hisar				6.87
8	Improvement work of Cooling Towers of RGTPP Unit I & II	6			
9	Replacement of 2 Nos. Stator of BCP of RGTPP Unit I & II		3.16		
10	Up gradation of C&I system for RGTPP Hisar	3	8.0		
11	Procurement of ID fan blades, RGTPP	1.65			
12	Replacement of 03 Nos. Fire Tenders at RGTPP		1.20		
13	Up gradation of hardware and software of PLC at RGTPP, Khedar, Hisar	4	3		

14	Replacement of 2 Nos. (one for each unit) Battery Banks for main plant 2x150 kVA UPS System for Unit 1 &2, RGTPP, Khedar, Hisar	0.41			
15	Procurement of Complete Battery Banks Lead Acid Plante 220V, 2140AH in each Unit (Unit 1&2), RGTPP, Khedar, Hisar		3.80		
16	Work for Supply, Erection, Testing and Commissioning of 02 Nos. ABB make unitrol-6080 Digital Automatic Voltage Regulator (DAVR) for Generator Excitation System and replacement with existing ABB make Unitrol-F DAVR at RGTPP, Khedar, Hisar			1.01	
17	Construction of First Aid Centre and additional RCC Roof slab of DG Set house at RGTPP, Khedar, Hisar	0.55			
18	Purchase of Ion Chromatography system fully automatic PC based, RGTPP Hisar	0.65			
19	Revival of Fire Fighting System of Unit6, PTPS, Panipat			0.60	
20	Replacement of damaged floor and Construction of Roads in PTPS Colony, Panipat as per new norms of Government of Haryana			1.55	
21	Up-gradation of PTPS Unit-6 HMI System of pro-control supplied by M/s BHEL			1.50	
22	Energy Management System PTPS Unit- 7-8		0.70		
23	Modernization of Boiler Lift for PTPS Unit 8		0.7		
24	Replacement of 02 Nos. Fire Tenders at PTPS Panipat	0.433		0.44	
25	Renovation of centralized AC System of Unit-7&8, PTPS Panipat		1.80		
26	Providing rejected Coal (Pucca Floor under 132 KV & 220 KV Lines inside the plant boundary) PTPS,Panipat	0.38			
27	Construction of all-weather patrolling track along the peripheral boundary wall at PTPS, Panipat	0.38			
28	Replacement of 8" water lines around the circular road in PTPS, Colony	0.50			
29	Township for DCRTTP, Yamunanagar		0.35	2.01	
30	Up-gradation of existing PLC & SCADA at DCRTTP		2.5488		
31	Refurbishment of BFP Cartridge: DCRTTP, YamunaNagar	1.55			
32	Purchase of 01 no. Runner Hub without blades and new set of guide vanes-WYC		7		
	TOTAL	26.16	77.19	11.54	11.87
	GRAND TOTAL	126.76			

Note:

1. The scheme listed at S.No 5, 01No of barrack has been completed and the same is sufficient for handling the work.
2. The scheme listed at S.No 6 the work is under progress and as per the work order the same has been revised downward from 2.78Cr to 2.44Cr.
3. The scheme at S.No 14 the amount of work is including the buyback of old batteries, thus the awarded cost considering the buyback has been reduced from 0.60Cr to 0.41Cr.

4. The scheme listed at S. No 26 & 27, the actual cost of work order as been reflected for approval and the same has been 1Cr approved vs 0.76 Cr as per actual.

Table 16 : Completed/Dropped Schemes

Sr. No.	Capital Expenditure Work	Amount	Remarks
1	Procurement of PA fan blades for RGTPP Hisar	0.42	Completed
2	Procurement of 2 No. Air Driers for Transport Compressors for RGTPP Hisar	0.75	Dropped
3	Revival of 02 Nos of ESP fields of RGTPP Unit I	6.17	Completed
4	Installation of Variable Frequency Drive in Condensate Extraction Pump (CEP) of RGTPP Unit I& II	5.21	Dropped
5	Replacement of PTPS Unit-6 AD Line in Ash Handling & repair D2 of ESP Field	0.35	Completed
6	Up gradation of existing DCS system for DCRTTP 1 & 2	0.35	Completed
7	Providing of 2 No. VFD on Unit-II DCRTTP ,6.6KV Motor of CEP	2.00	Dropped

Note :

CAPEX in respect of New Environmental Norms: The CAPEX in respect of implementation of the stringent New Environmental norms is yet to be finalised after selecting the best suitable option. HPGCL has already initiated the process for finalising the same after in principle approval accorded by the Hon'ble Commission. HPGCL will approach the Hon'ble Commission with its actual expenditure after completion/COD of the CAPEX in respect of the New Environmental Norms for approval. The indicative values exclusive of IDC & IEDC in respect of the New Environmental Norms CAPEX is as under:

Table 17: Indicative CAPEX for New Environmental Norms (Cr.)

Sr. No.	Capital Expenditure Work	2021-22	2022-23	2023-24	2024-25	Total
1	Installation of FGD RGTPP	66.55	399.31	133.11	66.55	665.52
2	Installation of FGD DCRTTP	55.14	330.86	110.29	55.14	551.44
3	Installation of FGD PTPS 6	6.04	34.23	-	-	40.27
4	Installation of FGD PTPS 7-8	13.55	60.79	-	-	74.34
5	Installation of Low NOx Burner &SOFA RGTPP	7.28	65.48	-	-	72.75
6	Installation of Low NOx Burner &SOFA DCRTTP					
7	Installation of Low NOx Burner &SOFA PTPS 7-8					

7.4 Hon'ble Commission is requested to approve the revised capitalization of schemes as per the details provided in table 15.

A8 TRUE-UP FOR FY 2020-21

Introduction

- 8.1 Generation tariff for FY 2020-21 was determined by the Hon'ble Commission as per HERC MYT Regulation, 2019 on the tariff Petition of HPGCL, vide its Order dated 24/04/20 read with order dated 28/05/2020.
- 8.2 As per the regulation 13.1 of the HERC, MYT Regulation, 2019, truing up of ARR of the previous year shall be carried out with midyear performance review of each year of the control period only when the audited accounts in respect of the year under consideration is submitted along-with the application. In case audited accounts pertaining to the year, of which truing up is to be undertaken, are not available, the generating company or the licensee as the case may be shall submit the provisional account duly approved by the BoD's of the company/ licensee.
- 8.3 Accordingly, HPGCL is submitting the true-up for FY 2020-21 based upon the audited accounts (**Annexure- P7**) for the FY 2020-21 for the approval of the Hon'ble Commission.
- 8.4 The following components have been considered by HPGCL for true up claim for FY 2020-21:-
- i) Operation & Maintenance (O&M) Expenses
 - ii) Depreciation
 - iii) Interest & Finance charges
 - iv) Interest on working capital
 - v) Return on Equity (RoE)
 - vi) Cost of Oil
- 8.5 Component wise variation in the approved fixed cost as adjudicated by HERC in terms of MYT Regulation, Recovered cost as per MYT regulation and actual fixed cost exclusive of solar business as per the audited accounts is submitted for consideration and approval of Hon'ble Commission for true-up of FY 2020-21.
- 8.6 The reference is invited to HERC MYT Regulation 2019 as under:
- "30. RECOVERY OF ANNUAL FIXED CHARGES (CAPACITY) CHARGES FOR THERMAL POWER PROJECTS*
- (a) The fixed cost of a thermal generating station shall be computed on annual basis, based on norms specified under these Regulations. Payment of capacity charge by the beneficiaries shall be on monthly basis in proportion to allocated / contracted capacity. The total capacity charges payable for a generating plant shall be shared by its*

beneficiaries as per their respective percentage share / allocation in the capacity of the generating plant;

(b) A generating plant shall recover full capacity charge at the normative annual plant availability factor specified by the Commission. Recovery of capacity charge below the level of target availability shall be on pro-rata basis.....”

8.7 Further the reference is invited to the Regulation 13 of MYT regulation 2019 as under:

“13. TRUING-UP

13.1 Truing-up of the ARR of the previous year shall be carried out along with mid-year performance review of each year of the control period only when the audited accounts in respect of the year(s) under consideration is submitted along with the application. In case audited accounts pertaining to the year, of which truing-up is to be undertaken, are not available, the generating company or the licensee as the case may be, shall submit the provisional account duly approved by the Board of Directors of the company/licensee.

13.2 Truing-up of uncontrollable items shall be carried out at the end of each year of the control period through tariff resetting for the ensuing year and for controllable items shall be done only on account of force majeure conditions and for variations attributable to uncontrollable factors.

13.3 The Commission shall allow carrying costs for the trued-up amount (positive or negative) at the interest rates specified in these Regulations by adjusting the interest allowed on the working capital requirement for the relevant year of the control period. Upon completion of the mid-year performance review and truing up in accordance with these regulations, the Commission shall pass an order recording:

(a)The revised ARR for such financial year including approved modifications, if any;

(b)Holding cost for under/over recovered amount from the close of the relevant year and upto the middle of the ensuing year of the control period whereupon the trued-up amount has been adjusted by appropriate resetting of tariff in accordance with regulation 13.4, calculated as additional borrowing for working capital for that period.

Provided that no carrying cost shall be allowed on account of delay in filing for true-up due to unavailability of the audited accounts

13.4 Over or under recoveries of trued-up amount in previous year(s) of the control period shall be allowed to be adjusted in the ensuing year of the control period by appropriate resetting of tariff. The unrecovered amount in the one control period shall be adjusted in the subsequent control period.”

The above regulations clearly specify the methodology for recovery of the fixed charges components as per the availability of the plants and the True-up is also required to be between “Recovered” as per plant availability and Actual as per audited accounts.

Thus, in view of the above regulations of MYT 2019, HPGCL has “Recovered” the components of the tariff as per the availability of the Units and accordingly, the “Recovered” shall be come the approved charges as per the regulations. Therefore, the True-up has been offered on “Recovered” vs Actual as per audited accounts in the matter.

Operation and Maintenance Expenses:

- 8.8 The operation and maintenance expense include actual expenses incurred by HPGCL towards Employees cost, Repair & Maintenance expenses and A&G expenses.
- 8.9 Employee expenses mainly comprise of salaries, dearness allowance, terminal benefits in the form of pension & gratuity, leave encashment and staff welfare expenses etc. Repair & Maintenance expenses are the expenses incurred towards day to day upkeep of generation assets for better and reliable operation. Administration & General expenses mainly comprise of property and material related expenses, communication expenses, professional charges, conveyance and travelling allowances, etc.
- 8.10 Hon'ble Commission had approved O&M Expenses of Rs. 788.79Cr for FY 2020-21. The total actual O&M Expenses as per audited accounts for FY 2020-21 remain Rs. 875.63Cr. The expenses relating to the net Actual O&M expenses (exclusive of solar business of Rs 0.62Cr) remain Rs. 875.01 Cr.

Table 18: Operation and Maintenance Expenses (Rs. Cr.)

Particular	FY 2020-21 (Approved) 1	FY 2020-21 (Recovered) 2	FY 2020-21 (Actual) 3	True-Up 4=(3-2)
Employee Cost	591.175	545.23	637.86	92.63
Repair & Maintenance	173.133	161.93	191.88	29.95
Administrative & General	24.483	22.53	45.27	22.74
Total	788.791	729.70	875.01	145.31

- 8.11 The primary reason for the significant gap between the approved, Recovered and actual O&M expenses is due to increase in uncontrollable expenses on account of the employee cost including terminal liabilities & non availability of RGTPP-2 since 19th September,2020 .
- 8.12 Approved Employees Cost for FY 2020-21 was Rs. 591.175 Crore as allowed by Hon'ble Commission, whereas, The actual employee cost for FY 2020-21 as per audited accounts remains Rs.637.86Crore inclusive of net Defined Benefit liability (terminal liability) of 402.08 Crore in respect of employees as per the Actuary Valuation Report (**Annexure- P8**) submitted by the independent actuary by M/s Kapadia Global Actuaries, Mumbai.

- 8.13 For the period in consideration i.e. FY 2020-21, HPGCL appointed M/s Kapadia Global Actuaries, Mumbai as independent valuer for the valuation for FY-2020-21. It is further submitted that valuation of the actuary is based on the given data and circumstances/ scenario and other market conditions for investment. HPGCL is having two employee welfare trusts namely HPGCL Employees Pension Fund Trust and HPGCL Employees Provident Fund Trust. Retirement benefit like Pension/ Gratuity Leave encashment and Commutation is being paid by HPGCL through its HPGCL Employees Pension Fund Trust. Trust makes the investment of the funds contributed by HPGCL based on the valuation of Actuary report according to Ministry of Finance, GOI, Department of Finance O/o no F.no.11/14/2013 dated 02.03.2015. Actuary valuation is based on various data like employees count/ average age of active employees/ number of pensioners/ average age of pensioners /monthly salary eligible for LTC (active employees &pensioners). Besides this information Actuary also considers Discount rate/salary escalation rate/ Attrition rate/ Withdrawal rate/ Mortality rate etc. Based on all the above information, an Actuary gives its report to Trust for safeguard the interest of the employees of HPGCL after retirement. The past year Trend of Terminal Liability of HPGCL evaluated by the independent Actuary from FY 2013-14 is as under:

S. No.	Financial Year	Amount in Cr.
1	2013-14	152.36
2	2014-15	250.76
3	2015-16	132.51
4	2016-17	478.07
5	2017-18	485.01
6	2018-19	688.45
7	2019-20	356.68
8	2020-21	402.08

Further, HPGCL is bound by the rules and regulations of State Government pertaining to employee's benefits (pay structure, D.A., annual increment@3%). Any revision, therefore in the pay structure of its employees is beyond the control of the HPGCL and falls under Regulation 8.3.8(b). All these factors lead to the increase in the employees cost of HPGCL.

- 8.14 Terminal liability is an uncontrollable expenditure under Regulation 8.3(b) of MYT Regulation 2019 and the same is being allowed by the Hon'ble Commission with the True-up of respective year.

Therefore in view of the above, the Hon'ble Commission is requested to kindly take note of the above and allow the net true up of Rs. 92.63 Cr. in respect of Employee Cost as per actual basis.

R&M and A&G Expenses:

- 8.15 The O&M expenses other than employee cost, i.e. R&M and A&G expenses approved by the Hon'ble Commission for FY 2020-21 are Rs 173.133 Crore & Rs 24.483 Crore. However, the amount recovered by HPGCL is Rs. 161.93Cr and 22.53Cr respectively due to reduced availability of HPGCL Units on account of non availability of RGTPP Unit-2 from September 19th, 2020 onwards . The actual R&M and A&G expense for the year remains Rs. 191.88Cr (excluding solar business) and Rs 45.27Cr.
- 8.16 The increase in R&M amount is primarily attributable to the cost of R&M being carried for RGTPP-2 and also on account of reduced R&M to 50% for Unit-6, PTPS allowed by Hon'ble Commission under its order dated 24.04.2020.
- 8.17 Further, the Hon'ble Commission in its order dated 18.02.2021 has identified that the donation contributed to the CM Corona Relief Fund on account of outbreak of pandemic COVID 19 Corona Relief Fund is pass through in nature. On similar lines, HPGCL has also contributed to CM Corona Relief Fund amounting to Rs 21.00 Cr which is the sole reason for increase in the A&G expenses. The same needs to be pass through in lines with the Hon'ble Commission order dated 18.02.2021
- 8.18 In view of above, Hon'ble Commission is requested to allow the true up of the O&M cost amounting to Rs. 145.31 Cr .i.e. The difference between the Recovered and Actual O&M cost as per audited accounts for FY 2020-21.

Depreciation

- 8.19 The actual depreciation of HPGCL for FY 2020-21 as per the audited accounts is Rs. 336.58Cr. This includes depreciation for solar assets to the tune of Rs. 3.18 Cr. Therefore, the depreciation exclusive of the solar business worked out as Rs 333.40Cr.
- 8.20 Further Hon'ble Commission in its Order dated 31.10.2018 & 07.03.2019 has directed HPGCL not to claim depreciation on spares and dismantling cost on account of Ind AS. Depreciation on Capitalization of spares and decommissioning cost for FY 2020-21 in accordance Ind AS, is Rs. 5.76 Cr & Rs. 11.41Cr. Thus, HPGCL in compliance with aforesaid directives has excluded a sum up to Rs 17.17Cr. (5.76+11.41) from its true up claim of Depreciation.
- 8.21 Accordingly, the net allowable Depreciation for FY 2020-21 exclusive of Solar business and depreciation on spares and decommissioning cost worked out as Rs. 316.23Cr (333.40-5.76-11.41). The approved depreciation for FY 2020-21 was Rs. 332.85 Cr.
- 8.22 The depreciation detail for the FY 2020-21 is as under:

Table 19: True-up of Depreciation (in Rs. Cr.)

S. No	Unit	Approved	Actual as per audited accounts*	Dep. on GAAP Spares	Dep. on account of Ind AS	Net allowable dep.	Recovered Dep.	Variance
A	B	C	D	E	F	G=(D-E-F)	H	I=(G-H)
1	PTPS-6	6.690	1.86	0.08	1.31	0.47	6.69	(6.22)
2	PTPS-7-8	58.45	57.03	0.47	5.01	51.55	58.45	(6.90)
3	DCRTPP	54.28	57.09	1.22	1.42	54.45	54.28	0.17
4	RGTPP	204.15	212.17	3.99	3.67	204.51	158.42	46.09
5	Hydel	9.28	5.25	-	-	5.25	9.00	(3.75)
	Total	332.85	333.40	5.76	11.41	316.23	286.84	29.39

*Excluding Solar business of Rs. 3.18Cr.

- 8.23 HPGCL falls short of Rs 29.39 Crore against the actual depreciation of the HPGCL Units. It is worth to intimate that the shortfall in recovery of tariff in respect of RGTPP-2 is the main reason for HPGCL to have short fall in recovery of allowed depreciation.

The Commission is requested to take note of above and allow recovering the depreciation under True up as proposed above.

- 8.24 A copy of the Fixed Assets Register for the FY 2020-21 has already been submitted to the Hon'ble Commission vide HPGCL memo 207 /HPGCL/REG-522 dated 09.11.2021 for kind consideration of the Hon'ble Commission.

Interest & Finance Charges

- 8.25 Hon'ble Commission in its order dated 24.04.2020 had approved the interest and finance charges for FY 2020-21 as Rs. 86.24 Cr with the condition that the interest on term loan has been taken as per the existing loan profile of HPGCL i.e. post restructuring, which shall be subject to true-up. HPGCL is proposing to claim the 50% of the savings (Rs. 143.10 Cr) to the beneficiary due to restructuring as per Regulation 21.1 (v) of HERC MYT Regulation, 2019.
- 8.26 Actual Interest and Finance Charges for FY 2020-21 as per audited account is Rs. 49.41Cr. An amount of Rs.2.31Cr. of the Interest and Finance Charges pertains to the Solar Business, as such net actual Interest and Finance Charges exclusive of solar business as per the audited accounts remains Rs. 47.11 Cr.
- 8.27 The saving in the Interest and Finance Charges is due to restructuring of its loan portfolio & advance Payments made by HPGCL by applying prudent financial management. Refinancing cost of such restructuring has already been allowed and adjusted by the Hon'ble Commission in its previous Tariff Orders and also identified the same in order dated 18.02.2021.
- 8.28 As per the regulation 21.1 (v) of HERC MYT Regulation, 2019 the cost associated with the refinancing has to be borne by the beneficiaries and the net savings after deducting the cost of refinancing shall be subject to incentive and penalty framework as per Regulation 12.

HPGCL has taken loans from various financial institutions for setting up of its power plants & related capex requirements. The loans originally taken from Power Finance Corporation, Rs. 965.48 Crore for RGTPP Hisar and Rs. 874.58 Crore for DCRTTP, Yamunanagar, were swapped through SBI in Feb 2016 & in April 2017 respectively. Interest and Finance charges for FY 2020-21 as per pre & post -restructuring Loan portfolio exclusive solar business are given below:

Table 20: Pre-Restructuring Loan Portfolio & Repayments schedule (Rs. Cr.)

Particulars	Rate of Interest	Opening Bal	Drawls during the year	Repayme nts during the year	Closing Balance	Interest during the year
GPF Bonds	7.10%	40.69	0.00	6.78	33.91	2.41
SBI DCRTTP YNR	12.50%	512.66	0.00	120.64	392.02	56.54
REC	12.25%	501.64	0.00	75.60	426.04	56.82
State Bank of India(RGTPP)	11.45%	539.94	0.00	101.64	438.30	56.00
APDP Loan	12.50%	3.11	0.00	0.15	2.96	0.38
Punjab National Bank (Andhra Takeover)	8.65%	21.05	0.00	20.00	1.05	0.96
Punjab National Bank (Andhra Takeover Hisar)	8.65%	57.40	0.00	38.00	19.40	3.32

Particulars	Rate of Interest	Opening Bal	Drawls during the year	Repayme nts during the year	Closing Balance	Interest during the year
Punjab National Bank (REC taken Over)	12.25%	122.77	0.00	20.52	102.25	13.78
NABARD	5.25%	45.99	0.00	11.50	34.49	2.11*
Total		1845.25	0.00	394.83	1450.42	192.33

* Total Interest during the year excluding solar business 192.33- 2.11= 190.22Cr

Table 21: Post -Restructuring (Actual) Loan Portfolio and Int. & Fin. Charges (Rs. Cr.)

Particulars	Rate of Interest (%)	Opening Bal	Additions during the year	Repayments during the year	Closing Balance	Interest during the year
GPF Bonds	7.10	40.69	-	6.78	33.91	2.41
SBI (DCRTPP)	-	-	-	-	0	0
REC	7.75	453.53	-	75.6	377.93	34.69
SBI(RGTPP)	-	-	-	-	0	0
APDP Loan	12.50	3.11	-	0.15	2.96	0.39
PNB(Andhra Takeover)	8.50	22.20	-	22.20	0	0.92
PNB(Andhra Takeover, Hisar)	8.50	56.93	-	56.93	0	2.47
PNB Loan	8.50	122.77	-	122.77	0	5.95
NABARD	5.25%	45.99		11.50	34.49	2.31*
PNB (SBI takeover)		21.14	-	21.14	0	0.28
Total		766.36	0	317.07	449.22	49.42

*solar business

8.29 HPGCL has paid compensation amounting to Rs. 7.30 Cr. to the land owners of RGTPP, Hisar in compliance to order of Hon'ble Supreme Court and Rs. 0.46 Cr. to the land owners of PTPS, Panipat in compliance of Hon'ble Punjab & Haryana High Court. The entire compensation is a capital expenditure of HPGCL and has been entirely funded by the State Govt. as equity.

8.30 As per Regulation 19.2 (b) of the HERC MYT Regulations 2019, the capital expenditure is to be funded in the Debt Equity ratio of 70:30. Equity in access of 30% would be treated as normative loan/ debt for the purpose of tariff determination and true-up.

8.31 Accordingly, HPGCL has considered Rs. 5.43Cr being 70% of the capital expenditure incurred on the land compensation of Rs. 7.76Cr(7.30+0.46) as normative debt at 8.5% rate of interest (average actual rate of interest of HPGCL). The normative interest expense so incurred stands at Rs 0.23 Cr. The same has been added to the final true-up of FY 20-21. The same was approved by the Hon'ble Commission vide its order dated 18.02.2021.

8.32 As per the approved Tariff order dated 24.04.2020, HPGCL is proposing to pass on 50% of the savings on interest and finance charges to the beneficiaries on account of restructuring of loan and consider the true up of interest & finance charges as given in the below table:

Table 22: Interest & Finance Charges (Rs. Cr.)

Particular	Approved interest & Finance Charges	Actual interest & Finance Charges	Pre-restructuring interest & Finance Charges	Savings on account of restructuring of loan	Allowable interest & Finance Charges	Recovered by HPGCL	True-up
1	2	3	4	5= 4-3	6=3+50%(5)	7	8=6-7
Int.&Fin. Charges (A)	86.24	47.11	190.22	143.10	118.66	70.12	48.54
Int. On Normative Debt(B)					0.23		0.23
Total True up of Int.& Fin. Charges(A+B)	86.24	47.11	190.22	143.10	118.89	70.12	48.77

8.33 The Hon'ble Commission is hereby requested to kindly approve the true-up of Rs 48.77Cr on account of interest and Finance Charges.

Return on Equity

8.34 As per regulation 20.1 of the HERC MYT Regulation 2019, Hon'ble Commission may allow the RoE up to 14%.

The Hon'ble Commission vide its order dated 24.04.2020 (page 84 under Note-3) has adjudicated the issues as under:

"Note – 3: As per the MYT Regulations, 2019, ROE upto 14% can be allowed on the eligible Equity Capital in use. The Commission, taking a holistic view of the power sector in Haryana including the tariff payable by the electricity consumers had traditionally restricted the RoE to 10% in order to cushion the tariff shock to the consumers. In FY 2020-21, given the unprecedented situation emanating from the COVID-19 pandemic and the resulting restriction/lockdown ordered by Central Government/State Government, all economic activities came to a standstill. Resulting, the ability to pay of all categories of consumers has been significantly reduced. The pandemic has impacted income, earnings and employment of all categories of consumers be it domestic consumers, industrial/commercial consumers. As far as Government connections are concerned i.e. Public Water Works, Street Light, Lift Irrigation as well as general connections in Government Offices and building, it is also a fact that due to significant reduction in revenue from direct/indirect taxes and levies, their ability to pay, has also been impaired. Hence, the Commission, after due deliberations, has considered not to allow any RoE in the FY 2020-21."

8.35 It is worth to apprise that HPGCL has opted for an appeal at Hon'ble APTEL against the issue of disallowance of RoE vides DFR no 216/2020. The same is pending for adjudication at APTEL.

8.36 Keeping in view of the above, the details of opening equity, equity addition and required return of equity considered unit-wise for FY 2020-21 is summarized in the table below for intimation only:

Table 23: Details of Equity Employed (Rs. Cr.)

Plants	Opening	Additions	Closing	RoE Recovered
PTPS – 6	156.77	0.07	156.84	0
PTPS – 7	218.04	-	218.04	0
PTPS – 8	218.02	-	218.02	0
DCRTPP-1	250.765	0.40	251.165	0
DCRTPP-2	250.715	0.40	251.115	0
RGTPP-1	493.3735	1.31	494.6835	0
RGTPP-2	492.8135	0.16	492.9735	0
Hydel	18.345	0.01	18.355	0
Total	2098.842	2.35	2101.192	0

8.37 HPGCL shall reserve its right for reconciliation/ True-up of RoE as per the outcome of the legal recourse opted in the matter.

Table 24: True-up of RoE (Rs Cr.)

Approved RoE (A)	Actual RoE (B)	Recovered RoE (C)	True-up of RoE Cost (B-C)
-	210.00	0	**

**matter pending at APTEL

Cost of Oil

8.38 As per the Regulation 31 of the MYT Regulation 2019 the Specific Oil Consumption becomes the part of the ECR as under

"31 ENERGY CHARGES OR VARIABLE CHARGES FOR THERMAL POWER PROJECTS

(a) The Energy charges or variable charges shall cover the main fuel cost & secondary fuel oil and shall be payable for the total energy scheduled to be supplied to a beneficiary during the calendar month on ex-power plant basis, at the specified variable charge rate, with fuel price adjustment.

(b) The Energy charge for the month shall be worked out on the basis of exbus energy scheduled to be sent out from the generating plant in accordance with the following formula:

Energy charge or variable charge (Rs)

Energy Charge Rate (Rs. / kWh) x Scheduled Energy (ex-bus) for the month (kWh)

Note: Till the time intra state ABT is implemented, 'scheduled energy' may be read as 'actual energy sent'.

(c) Energy charge rate (ECR) in Rs. per kWh on ex-power plant basis shall be determined to three decimal places in accordance with the following formula:

(i) In case secondary fuel Oil cost is the part of ECR: $\frac{[SHR-(SFCXCVSF) \times LPPF]}{CVPF} + \frac{(SFCXLPSF)}{100} \times \frac{100}{(100-Aux)}$

(ii) In case secondary fuel Oil cost is not the part of ECR

$$\{[SHR-(SFC \times CVSF) \times LPPF]/CVPF\} \times \{100/(100-Aux)\}$$

Where

AUX = Normative auxiliary energy consumption in percentage;

CVPF = Gross calorific value of primary fuel as fired, in kCal per kg or per litre as applicable;

CVSF = Gross calorific value of secondary fuel in kCal per ml; ECR = Energy charge rate in Rs. per kWh sent out;

SHR = Normative Station Heat rate in kCal per kWh; SFC = Normative Specific fuel oil consumption in ml/kWh

LPPF = Weighted average landed price of primary fuel in Rs./kg.

LPSF = Weighted average landed fuel cost of Secondary Fuel in Rs./ml during the month"

8.39 Further, the reference is invited to Regulation 29 of MYT 2019, wherein it has been guided as under:

"29. EXPENSES ON SECONDARY FUEL OIL FOR THERMAL POWER PROJECTS

(c) Expenses on secondary fuel oil (in Rs.) shall be computed corresponding to normative secondary fuel oil consumption (SFC) specified in this Regulation, in accordance with the following formula:

$$\text{Expenses on secondary fuel oil (in Rs.)} = SFC \times LPSFi \times NPAF \times 24 \times NDY \times IC \times 10$$

Where, SFC = Normative specific fuel oil consumption in ml/kWh;

LPSFi = Weighted average landed price of secondary fuel in Rs. / ml considered initially;

NPAF = Normative annual plant availability factor in percentage;

NDY = Number of days in a Year;

IC = Installed capacity in MW

(b) Initially, the landed cost of secondary fuel oil shall be considered based on the weighted average price of secondary fuel oil during the three preceding months and in the absence of landed costs for the three preceding months, latest procurement price for the generating plant, before the start of the year shall be considered

(c) The secondary fuel oil expenses shall be subject to fuel price adjustment at the end of each year of tariff period as per following formula:

$$SFC \times NPAF \times 24 \times NDY \times IC \times 10 \times (LPSFy - LPSFi)$$

Where, LPSFy = Weighted average landed price of secondary fuel oil for the Year in Rs./ml"

8.40 It is worth to apprise that HPGCL PLF remains on lower side & higher Start Stop operations of the Units leads to higher Oil consumption which remains unrecovered. The other reason attributable to lower scheduling of power by DISCOMs, due to the reasons known to them. As per regulation 29, the Normative ceiling of Oil recovery is there, which allows HPGCL to recover the Oil cost as per the prevailing regulations. Thus, HPGCL proposes the true up of Oil Cost as under:

Table 25: Details Cost of Oil (Rs Cr.)

Allowed as per NAPAF	Actual	Recovered through ECR	True-up of Oil
47.541	15.27	14.23	1.04

8.41 Hon'ble Commission is requested to approve true-up of Rs 1.04 Cr on account of oil cost for FY 2020-21 as submitted above.

Interest on working Capital

8.42 Interest on working capital amounting to Rs 109.668Cr has been allowed by the Hon'ble Commission in its order dated 24.04.2020 regarding generation tariff for FY 2020-21 after considering the average coal and oil prices without any fuel price escalation as proposed by HPGCL. However, there has been variation in prices of coal and oil during the financial year in consideration for true-up. Therefore, while computing the true-up of working capital for FY 2020-21, actual rate of coal has been considered.

8.43 Due to variation in Fuel prices, the interest on normative working capital requirement for FY 2020-21, as per the approved norms of the HERC, has decreased to Rs 106.03Cr against the approved interest on working capital of Rs 109.668Cr.

As per HERC MYT regulation 2019, the Interest on working capital on normative basis is under:

"22. INTEREST ON WORKING CAPITAL

22.1 For the purpose of computing working capital the components mentioned in the table below shall be considered:

"I. Coal-based Thermal Generating Plants:

- a) Cost of coal for 01 months corresponding to the normative availability;*
- b) Cost of secondary fuel oil for 01 months corresponding to the normative availability;*
- c) Normative O&M expenses for 1 (one) month;*
- d) Maintenance spares @ 10% of the O&M expenses;*
- e) Receivables equivalent to fixed and variable charges for 1 (one) month for sale of electricity calculated corresponding to normative availability."*

However the HPGCL is claiming the true-up of IWC on the basis of MYT 2019 as under:

"12. INCENTIVE AND PENALTY FRAMEWORK

12.1 Various elements of the ARR of the generating company and the licensee will be subject to incentive and penalty framework as per the terms specified in this regulation. The overall aim is to incentivize better performance and penalize poor performance, with the base level as per the norms / benchmarks specified by the Commission.

12.2 The elements of ARR of generating company and licensees to which incentive and penalty framework shall apply are as follows:

(a) Common for generating company and licensees

(i) Operation & maintenance expenses-Applicable when the actual expenses fall below or exceed the level specified by the Commission.

(ii) Interest on new long-term loans- Applicable when interest rate falls below or exceeds the level specified by the Commission.

(iii) Restructuring of capital cost - Applicable when there is a benefit from restructuring of capital cost.

(iv) Interest on working capital- Applicable when interest rate falls below or exceeds the level specified by the Commission

(vi) Restructuring of loan portfolio- Applicable when there is a net benefit from restructuring of loan portfolio”

8.44 HERC has approved the Interest on Working Capital @8.65%, however, due to change in interest rate, the allowable IWC as per Regulation 22.2 of MYT 2019 has been reduced to 8.60% which is liable to be pass through under the True-up.

8.45 The below table summarizes True-up of interest on working capital for FY 2020-21.

Table 26: Detail of WC & IWC (Rs. Cr.)

S.No	Particular	Approved IWC (Rs. Cr) @ 8.65%	Normative(Revised) IWC (Rs. Cr) @ 8.60%
1	Coal Stock	503.084	485.76
2	Oil Stock	3.962	3.96
3	O&M Expenses	65.733	65.73
4	Maint. Spares	78.217	78.22
5	Receivables	616.841	599.21
6	Total W/C Requirement	1267.836	1232.88
	Int. on W/C	109.668	106.03

Table 27 : True-up of IWC (Rs Cr.)

Particular	Approved IWC @8.65% (A)	Normative IWC @ 8.60% (B)	Actual (C)	Recovered IWC (D)	True-Up E=C-D
Interest on WC	109.668	106.03	115.45	97.70	17.75

8.46 HPGCL requests the Hon’ble Commission, to allow Rs 17.75 Cr as True-up of interest on working capital for FY 2020-21.

Summary of the True-up

8.47 Based on above submission, HPGCL summaries proposed total true-up for FY 2020-21 as per below table:

Table 28: Total True-up (Rs. Cr.)

	O&M Expense	Depreciati on Cost	Oil Expense	IWC	Interest & Fin. Charges	RoE	Total True-up
Total	145.31	29.39	1.04	17.75	48.77	-*	242.26

*RoE is subject to true up after the outcome of the appeal filed by HPGCL in the matter at APTEL vide DFR No 216/2020

Non-Tariff Income

8.48 Detail of Other Non-operating income included in the other income as per the Audited Balance Sheet for the FY 2020-21 is as under:

Table29: Non-operating income (Rs. Cr.)

Particulars	Amount in Cr.
Income from sale of scrap	3.78
50% of Other income	1.61
Total	5.39

8.49 Net True-up for the FY 2020-21 after reducing the other non-operating income is given as under:

Table30: Net True-up after adjusting Non Tariff income (Rs. Cr.)

Particulars	Amount (Rs. in crore)
Total True up as per Table-27	242.26
Less: Non-operating income as per Table-28	5.39
Net True up	236.87

8.50 HPGCL requests Hon'ble Commission to approve True-up as submitted above for FY 2020-21 along with applicable carrying cost. The carrying cost should be allowed as per the regulations of MYT 2019 along with LPS, if any delay in realization of payments beyond the stipulated time on DISCOMs as per clause 43 of MYT Regulation 2019.

A9 MID-YEAR PERFORMANCE REVIEW FOR FY 2021-22 AND GENERATION TARIFF FOR FY 2022-23

- 9.1 HPGCL, in this section, submits the performance parameters for mid-year performance review for FY 2021-22 in line with Regulation 11 of the HERC MYT Regulations, 2019.
- 9.2 Hon'ble Commission, in its previous orders, has not carried out mid-year review. Accordingly, HPGCL has not submitted the tariff impact for the midyear performance review for FY 2021-22. However, performance indicators have been submitted for the year for kind consideration of the Hon'ble Commission.
- 9.3 As per MYT regulation 2019, the oil cost becomes the part of Energy Charge Rate (ECR) and has been calculated at normative basis. However, on account of low loading of the HPGCL units and frequent start & stop operations, the limits provided in the ECR as per regulation for specific oil consumption is on lower side at current PLF, HPGCL reserve its right to claim the under recovery of Oil cost in future true up as per Regulation 29 of MYT Regulation 2019.
- 9.4 HPGCL reserves its rights for making additional submissions at filing of true up, if required in the matter.

Technical Parameters

NAPAF

- 9.5 The Hon'ble Commission reference is invited to last 12 years data for WYC, wherein the actual CUF vs actual discharge is demonstrated as under:

Table31: Historical Actual CUF & Average Annual Discharge

Financial year	CUF (in %)	Average annual discharge from Hathnikund Barrage (in cusecs)
2009-10	43.50	2965
2010-11	49.85	3510
2011-12	52.20	3782
2012-13	45.22	3770
2013-14	37.33	4191
2014-15	32.58	3954
2015-16	34.47	3490
2016-17	37.55	3072
2017-18	32.30	3221
2018-19	43.48	3360
2019-20	54.89	4093
2020-21	44.61	3604
Average values	42.33	3584

- 9.6 The Hon'ble Commission may take the note of the above, and allow the actual CUF achievable in the matter for WYC Hydel @ 42.33% as per actual data and annual discharge of water.
- 9.7 Therefore, HPGCL proposes NAPAF for WYC @ 43% on the basis of last 12 years of data and on the basis of the water discharge for the FY 2022-23. The secondary energy, if any, shall be treated as per instant regulation.
- 9.8 It is worth to apprise the Hon'ble Commission that Hon'ble APTEL vides order dated 24.09.2021 has allowed the tariff to be recovered for Unit-6, PTPS as per the PPA including RoE. Thus, HPGCL is proposing the tariff for Unit-6, PTPS for future as per the instant regulations.
- 9.9 Therefore, HPGCL proposes the same NAPAF for thermal Units for FY 2021-22 in line with Tariff Order dated 18.02.2021 and for the FY 2022-23 in line with HERC MYT Regulation, 2019 as tabulated below:

Table32: NAPAF for FY 2021-22 and FY 2022-23

S.N	Unit #	Approved	Proposed	
		FY 21-22	FY21-22	FY 22-23
1	PTPS 6	35.00%	35.00%	85.00%
2	PTPS 7	53.00%	53.00%	85.00%
3	PTPS 8	53.00%	53.00%	85.00%
4	DCRTPP 1	85.00%	85.00%	85.00%
5	DCRTPP 2	85.00%	85.00%	85.00%
6	RGTPP 1	85.00%	85.00%	85.00%
7	RGTPP 2	85.00%	85.00%	85.00%
8	WYC Hydel	56.00%	56.00%	43.00%

Auxiliary Consumption

- 9.10 HPGCL hereby proposes the Auxiliary consumption for FY 2022-23 in line with MYT Regulations 2019 & for FY 2021-22 as already approved by the Hon'ble Commission is as tabulated below:

Table 33: Auxiliary Consumption (%) for FY 2021-22 and FY 2022-23

S.N	Unit #	Approved	Proposed	
		FY 21-22	FY21-22	FY 22-23
1	PTPS 6	9.00%	9.00%	10.00%
2	PTPS 7	8.50%	8.50%	9.30%
3	PTPS 8	8.50%	8.50%	9.30%
4	DCRTPP 1	8.50%	8.50%	8.50%
5	DCRTPP 2	8.50%	8.50%	8.50%
6	RGTPP 1	6.00%	6.00%	6.00%
7	RGTPP 2	6.00%	6.00%	6.00%
8	WYC Hydel	1.00%	1.00%	1.00%

- 9.11 As the PTPS Unit-5 & 6 were envisaged together and thus sharing some common auxiliaries. After the decommissioning of Unit-5, the Common auxiliaries are needed to be on bar for readiness of Unit 6. Thus, leads to higher auxiliary consumption for Unit 6. The Hon'ble Commission is requested to take the note of the above and allow the auxiliary @10% for PTPS Unit-6.

Secondary Fuel Consumption (SFC)

- 9.12 HPGCL proposes the Secondary Fuel Consumption for FY 2021-22 & FY 2022-23 as per HERC MYT Regulations, 2019.

Table 34: SFC (ml/kWh) for FY 2021-22 and FY 2022-23

S.N	Unit #	Approved	Proposed	
		FY 21-22	FY21-22	FY 22-23
1	PTPS 6	1.00	1.00	1.00
2	PTPS 7	0.50	0.50	0.50
3	PTPS 8	0.50	0.50	0.50
4	DCRTPP 1	0.50	0.50	0.50
5	DCRTPP 2	0.50	0.50	0.50
6	RGTPP 1	0.50	0.50	0.50
7	RGTPP 2	0.50	0.50	0.50

- 9.13 As per MYT regulation 2019, the oil cost becomes the part of Energy Charge Rate (ECR) and has been calculated on normative basis. However, on account of low loading/scheduling of the HPGCL units and frequent start & stop operations, the norms provided as per regulation for specific oil consumption is on lower side at current PLF/ loading of Units.
- 9.14 HPGCL reserve its right to claim the deficit in respect of the Oil consumption at the time of true-up of FY 2021-22 as per Regulation 29 of MYT Regulation, 2019.

Station Heat Rate (SHR)

- 9.15 The SHR for FY 2022-23 is being proposed as per norms specified in HERC MYT Regulations, 2019 as under:

Table 35: SHR (kCal/kWh) for FY 2021-22 and FY 2022-23

S.N	SHR(kcal/kWh)	Approved	Proposed	
		FY 21-22	FY 21-22	FY 22-23
1	PTPS 6	2550	2550	2550
2	PTPS 7	2500	2500	2500
3	PTPS 8	2500	2500	2500
4	DCRTPP 1	2344	2344	2344
5	DCRTPP 2	2344	2344	2344
6	RGTPP 1	2387	2387	2387
7	RGTPP 2	2387	2387	2387

Calorific Value and Price of fuel

9.16 The GCV and cost of coal& secondary fuel (oil) has been proposed for FY 2022-23 as per the actual weighted average of coal & landed of oil during April to Sept. of FY 2021-22 are as under:

Table 36: Proposed Gross Calorific Value & cost of coal for FY 2022-23

Particulars	PTPS	DCRTPP	RGTPP
Gross Calorific Value of Coal (kcal/Kg)	3439	3207	3368
Average landed cost of Coal (Rs/MT)	4570	4448	4797

Table 37: Proposed Gross Calorific Value & cost of oil for FY 2022-23

Particulars	PTPS	DCRTPP	RGTPP
Gross Calorific Value of oil (kcal/L)	9540	8994	9408
Average landed cost of oil (Rs/KL)	54451	51671	55456

Energy Charges

9.17 HPGCL is proposing for computation of ECR as per Regulation 31 of HERC MYT Regulation, 2019 as under:

Table 38: Computation of ECR for FY 2022-23

Unit	Generation (Ex-bus)	Per Unit Variable Cost
	MU	Rs/ Unit
PTPS 6	1407.29*	3.812*
PTPS 7	1688.38**	3.686**
PTPS 8	1688.38**	3.686**
DCRTPP 1	2043.93	3.574
DCRTPP 2	2043.93	3.574
RGTPP 1	4199.54	3.639
RGTPP 2	4199.54	3.639

*- The APC of Unit-6, PTPS has been considered @10% as proposed

** The APC of Unit-7&8, PTPS has been considered @ 9.3%(including additional auxiliary of 0.8% for Tube mill)

Annual Fixed Cost

9.18 HPGCL is projecting following components of fixed cost for FY 2021-22 as per HERC MYT Regulations, 2019.

O&M Expense

9.19 Hon'ble Commission has prescribed the norms for O&M Expenses (in Rs. Lac per MW) for FY 2022-23 under Regulation 28 of HERC MYT Regulations, 2019.

9.20 HPGCL has opted for an appeal at Hon'ble APTEL against the HERC order dated 24.04.2020 & 18.02.2021 against the reduction of Employee Cost from the base year.

9.21 HPGCL is claiming the Employee Cost as per regulation; however, reserve its right to revise the same as per the outcome of the appeal filed in Hon'ble APTEL, if required.

- 9.22 Considering above submissions HPGCL is proposing the O&M expense for FY 2022-23 as per the Hon'ble Commission in HERC MYT Regulation, 2019 is as under:

Table 39: O&M Expenses (Rs. Cr.)

S.N	Unit #	Approved	Proposed as per MYT,2019
		FY 21-22	FY 22-23
1	PTPS 6*	86.88	116.47
2	PTPS 7	88.873	128.50
3	PTPS 8	88.873	128.50
4	DCRTPP 1	111.566	137.85
5	DCRTPP 2	111.566	137.85
6	RGTPP 1	135.123	166.98
7	RGTPP 2	135.123	166.98
8	WYC Hydel	27.535	43.35
9	Total	785.538	1026.48

Note: Unit-6 as per interim stay of APTEL the tariff is being claimed as approved in FY 20-21 for FY 21-22 and for FY 22-23 Normative as per norms has been proposed.

- 9.23 Hon'ble Commission is requested to approve the O&M expenses for FY 2022-23 as Rs.1026.48 Crore.

Depreciation

- 9.24 Depreciation for FY2022-23 has been proposed on the basis of opening GFA as on 01.04.2022 and closing GFA as on 31.03.2023 as tabulated in Table 40 and after taking into consideration of the earlier disallowances made by the Hon'ble Commission in previous years and also with the additions of the CAPEX Schemes completed during the year as submitted in Chapter A7.
- 9.25 Opening Gross Fixed Assets for FY 2021-22 is as per the FAR of FY 2020-21 and Closing GFA for FY 2022-23 is after considering the additions of the CAPEX scheme completed in the respective years, as stated in Chapter A7, is tabulated below:

Table 40: Gross Fixed Assets (Rs. Cr.)

S.N	Unit #	GFA as on 01.04.20	Addition FY 2020-21	Addition FY 2021-22	Addition FY 2022-23	GFA as on 31.03.2023
1	PTPS – 6	996.82	0.00	-	-	996.82
2	PTPS – 7	945.18	0.34	0.85	8.74	955.10
3	PTPS – 8	954.99	0.22	0.85	9.44	965.48
4	DCRTPP-1	1,144.53	2.08	0.78	8.94	1,156.32
5	DCRTPP-2	1,141.89	2.07	0.78	8.94	1,153.68
6	RGTPP-1	2,181.22	8.33	9.71	17.07	2,216.32
7	RGTPP-2	2,173.75	0.96	9.71	17.07	2,201.48
8	Hydel	208.16	0.14	3.50	7.00	218.79
	Total	9746.52	14.13	26.16	77.19	9863.99

- 9.26 Hon'ble Commission in its earlier orders has disallowed certain capitalization, accordingly the allowable GFA is tabulated as under;

Table 41: Allowable GFA (Cr.)

Unit	GFA as on 01.04.2021 exclusive of Ind AS	Disallowances – GAAP spares	Allowable GFA as on 01.04.2021	Addition during 2021-22	Allowable GFA as on 01.04.2022	Addition during 2022-23	Allowable GFA as on 31.03.2023
PTPS – 6	996.82	1.07	995.75	-	995.75	-	995.75
PTPS – 7	945.52	2.35	943.17	0.85	944.02	8.74	952.76
PTPS -8	955.20	5.53	949.67	0.85	950.52	9.44	959.95
DCRTP-1	1,146.61	13.65	1,132.96	0.78	1,133.74	8.94	1,142.67
DCRTP-2	1,143.96	13.65	1,130.32	0.78	1,131.09	8.94	1,140.03
RGTPP-1	2,189.54	37.76	2,151.78	9.71	2,161.49	17.07	2,178.56
RGTPP-2	2,174.70	37.76	2,136.94	9.71	2,146.65	17.07	2,163.72
Hydel	208.29	-	208.29	3.50	211.79	7.00	218.79
Total	9760.64	111.76	9648.89	26.16	9,675.05	77.19	9752.24

9.27 The depreciation has been considered on the basis of rate as notified in HERC, MYT Regulations, 2019.

9.28 Further, the Hon'ble Commission in its Order dated 31.10.2018 & 07.03.2019 has directed HPGCL not to claim depreciation on such disallowed capitalization (spares and decommissioning cost).

9.29 Thus, HPGCL in compliance with aforesaid directives has excluded such depreciation of for the purpose of tariff determination of FY 2022-23 and the net allowable depreciation is as under:

Table 42: Depreciation (Rs. Cr.)

S.No.	Unit	Net allowable depreciation
1	PTPS - 6	0.46
2	PTPS - 7	26.50
3	PTPS - 8	27.45
4	DCRTPP-1	27.90
5	DCRTPP-2	28.23
6	RGTPP-1	103.35
7	RGTPP-2	103.19
8	Hydel	7.74
	Total	324.82

9.30 Hon'ble Commission in its Order dated 31.10.2018 had directed HPGCL to maintain a memorandum accounts of allowed capitalization, spares & decommissioning cost capitalized, depreciation and net block of fixed assets. HPGCL has maintained the memorandum accounts as directed by Hon'ble Commission and the same is attached as **Annexure-P9** for kind consideration of the Hon'ble Commission.

Interest & Finance Charges

- 9.31 Hon'ble Commission has approved the loan portfolio for HPGCL from time to time based on the approved CAPEX. Further, HPGCL by using its financial prudence has been successful in restructuring its loan portfolio to reduce the interest and finance charges.
- 9.32 The pre- restructuring and post restructuring loan portfolio of HPGCL as on 31.03.2020 have been presented in table 20 and 21 respectively.
- 9.33 HPGCL is expecting to incur interest and finance charges amounting to Rs 30.52 Cr. in FY 2022-23 while the pre restructuring interest and finance charges for FY 2022-23 are Rs 116.50 Cr. Therefore, there will be expected saving of Rs 85.98 Cr (Rs 116.50-30.52) in the interest and finance charges due to diligence and efficient financial management of HPGCL.
- 9.34 According to Clause 21.1 (v) of the HERC MYT Regulations 2019, HPGCL is eligible for incentive on the **net savings** resulting from restructuring of loan. Accordingly, HPGCL requests the Hon'ble Commission to approve interest expenses including incentive (50% of savings from restructuring) for FY 2022-23.
- 9.35 As per Para 8.29 to 8.31 above an amount of Rs. 5.43 crore of the equity contribution has been considered as normative debt @ 8.5% as per Regulation 19.2(b) of the HERC MYT Regulations 2019. The normative interest expense so incurred stands at Rs 0.23 Cr. The same has been added to the interest and finance charges for tariff computation of FY 2022-23.
- 9.36 The interest and finance charges for the FY 2022-23 so computed based on the above submissions are presented below;

Table 43: Interest and finance charges (Rs. Cr.)

	Int. & Fin. Charges post restructuring	Int. & Fin. Charges pre restructuring	Savings due to restructuring	Incentive (50% of savings)	Total interest expense	Interest expense on normative loan, if any	Final Interest Expense
1	2	3	4= (3-2)	5=50% of 4	6= (2+5)	7	8=6+7
PTPS 6	-	-	-	-	-	-	-
PTPS 7	0.92	0.99	0.07	0.03	0.96	0.01	0.96
PTPS 8	0.94	1.01	0.07	0.03	0.97	0.01	0.98
DCRTPP-1	0.97	14.19	13.22	6.61	7.58	0.00	7.58
DCRTPP-2	0.97	14.19	13.22	6.61	7.58	0.00	7.58
RGTPP-1	12.95	42.64	29.70	14.85	27.79	0.11	27.90
RGTPP-2	12.95	42.64	29.70	14.85	27.79	0.11	27.90
WYC Hydel	0.84	0.84	0.00	0.00	0.84	0.00	0.84

	Int. & Fin. Charges post restructuring	Int. & Fin. Charges pre restructuring	Savings due to restructuring	Incentive (50% of savings)	Total interest expense	Interest expense on normative loan, if any	Final Interest Expense
Total	30.52	116.50	85.98	42.99	73.51	0.23	73.74

9.37 Hon'ble Commission is requested to approve the same as interest and financial charges for FY 2022-23.

Return on Equity

9.38 Hon'ble HERC in its order dated 18.02.2021 has approved the RoE at 10%. However, Regulation 20 of HERC MYT Regulations, 2019, specifies the provision of Return on Equity capital at a ceiling of 14% per annum on the opening equity base of the particular year and also on 50% of allowable capital cost for the assets put to use during the year. Accordingly, HPGCL has considered Return on Equity at 14 % in line with the HERC Tariff Regulations 2019.

9.39 Accordingly, the equity employed and RoE for FY 2022-23 as per Regulation 20 of HERC MYT 2019 is tabulated below:

Table 44: Equity Employed (Rs Cr.)

S.N	Unit#	Closing FY 2020-21	Additions FY 2021-22	Additions FY 2022-23	Closing FY 2022-23	Proposed RoE @ 14%
1	PTPS – 6	156.84	-	-	156.84	21.96
2	PTPS – 7	218.04	0.17	1.75	219.96	30.67
3	PTPS – 8	218.02	0.17	1.89	220.08	30.68
4	DCRTPP-1	251.16	0.16	1.79	253.10	35.31
5	DCRTPP-2	251.11	0.16	1.79	253.05	35.30
6	RGTPP-1	494.69	1.94	3.41	500.04	69.77
7	RGTPP-2	492.97	1.94	3.41	498.33	69.53
8	Hydel	18.36	0.70	1.40	20.46	2.77
	Total	2101.19	5.23	15.44	2121.86	295.98

Interest on Working Capital (IWC)

9.40 Regulation 22.1 of HERC MYT Regulations, 2019 lists the components of working capital to be considered for estimating tariff. Further, Regulation 22.2 of the aforementioned Regulations states that the rate of interest on working capital shall be equal to the MCLR of the relevant financial year plus a maximum of 150 basis points.

9.41 SBI MCLR as on 01.04.2021 was 7.00% p.a. Accordingly, HPGCL has estimated the working capital requirements and the interest on working capital @ 8.50% (7.00%+1.50%) as per Regulation 22.2 of HERC MYT Regulation, 2019.

9.42 HPGCL is presently proposing the IWC as per MYT, 2019 regulations for FY 2022-23 is tabulated below.

Table 45: Calculation of IWC (Normative) (Rs Cr.)

Unit #	Coal Stock 1 Month	Oil Stock 1Month	O&M Expenses 1Month	Maint. Spares 10% Thermal/ 7.5% (Hydel)	Receivables 1 Months	Total W/C Requirement	Int. on W/C 8.50%
PTPS – 6	43.99	0.71	9.71	11.65	57.15	123.20	10.47
PTPS – 7	51.44	0.42	10.71	12.85	68.43	143.85	12.23
PTPS – 8	51.44	0.42	10.71	12.85	68.51	143.93	12.23
DCRTP-1	60.40	0.52	11.49	13.79	79.44	165.63	14.08
DCRTP-2	60.40	0.52	11.49	13.79	79.46	165.66	14.08
RGTPP-1	126.32	0.96	13.92	16.70	160.28	318.18	27.04
RGTPP-2	126.32	0.96	13.92	16.70	160.24	318.14	27.04
Hydel			3.61	3.25	4.64	11.50	0.98
Total	520.31	4.51	85.54	101.56	678.15	1390.09	118.16

Total Fixed Cost

9.43 Based on above submissions total fixed cost of HPGCL Plants for FY 2022-23 is tabulated below:

Table 46: Annual Fixed Cost (Rs. Cr.)

S.N	Unit #	O&M	Depreciation	Interest & Finance Charges	Return on Equity	W/C Interest	Total Fixed Cost
1	PTPS - 6	116.47	0.46	-	21.96	10.472	149.36
2	PTPS - 7	128.50	26.5	0.96	30.67	12.227	198.86
3	PTPS - 8	128.50	27.45	0.98	30.68	12.234	199.84
4	DCRTPP-1	137.85	27.9	7.58	35.31	14.078	222.72
5	DCRTPP-2	137.85	28.23	7.58	35.3	14.080	223.04
6	RGTPP-1	166.98	103.35	27.9	69.77	27.045	395.04
7	RGTPP-2	166.98	103.19	27.9	69.53	27.042	394.64
8	Hydel	43.35	7.74	0.84	2.77	0.978	55.68
	Total	1026.476	324.82	73.74	295.99	118.16	1839.18

9.44 HPGCL requests the recovery of all expenditure relating to petition filing fees including publication of notices etc. and any other statutory fees/ regulatory fees, taxes and levies from the beneficiaries as per actual.

9.45 Hon'ble Commission is further requested to allow Plant wise recovery of Fixed Cost on account of plant as sought in Chapter A5.4.

9.46 The petition has been prepared based upon the information available and to the best of our knowledge and belief of HPGCL. In case any information/clarification is required, the same may please be asked for, so that the same can be provided /clarified.

Summary of tariff computation for FY 2022-23

9.47 Based on above submissions the proposed tariff i.e. Total Capacity Charges and Energy Charge Rate (ECR) per kWh for FY 2022-23 is summarized as under:

Table 47: Tariff Summary for FY 2022-23

Particular	PTPS 6	PTPS 7	PTPS 8	DCRTP 1	DCRTP 2	RGTP 1	RGTP 2	WYC	Total
Total Capacity Charges (Rs Crore)	149.36	198.86	199.84	222.72	223.04	395.04	394.64	55.68	1839.18
Energy Charge Rate (Rs/kWh)	3.812	3.686	3.686	3.574	3.574	3.639	3.639		

A10 PRAYERS

10.1 HPGCL humbly prays Hon'ble Commission to:

- a) Admit this Petition.
- b) To direct DISCOMs to give minimum scheduling of quantum of power to HPGCL, as envisaged while deciding the APPC as submitted in Chapter A5.1.
- c) To direct DISCOMs to exhaust all the concluded contracts before resorting to Short Term purchase by considering all aspects as submitted in Chapter A5.2.
- d) To allow the recovery of fixed cost on plant basis by exercising Regulation 78 & 79 of the MYT, 2019, as all the regulations envisage metering on plant basis at outgoing feeder only, as submitted in Chapter-A5.3
- e) To relax the norms for Auxiliary Power Consumption for PTPS on account of Tube Mills at par with CERC Norms by exercising the regulation 78 & 79 of MYT Regulation 2019, as submitted in Chapter A5.4.
- f) To grant in-principle approval for increase in stock of Fuel to 60 days to meet the demand of Peak season by exercising the regulation 78 & 79 of MYT, Regulation, 2019 with the liberty to seek any increase in IWC at the time of true-up, as submitted in chapter A5.5.
- g) To consider MoEFCC notification as 'Change in law' and allow recovery of the expenditure incurred on account of transportation of Fly Ash under True-up on yearly basis, as submitted in Chapter: A5.6.
- h) Approve revised schedule of capital expenditure plan as submitted.
- i) Approve True-up of Rs 236.87 Cr. for FY 2020-21, after considering the Non Tariff Income as proposed as per audited financial statements with appropriate holding cost.
- j) Allow WYC PLF @ 43% on the basis of past trend of the generation and water flow data, as submitted in Chapter A9.
- k) Allow the normative parameters for PTPS for FY 2022-23 as per MYT, 2019 as proposed in chapter A9.
- l) Allow the Normative fixed cost and Normative ECR for FY 2022-23 as submitted in Chapter A9.

- m) Allow recovery of all expenditure relating to petition filing fees including publication of notices etc. and any other statutory fees/ regulatory fees, taxes and levies from the beneficiaries as per actual.
- n) Condone any inadvertent omissions / errors / delays / short comings and permit the applicant to add/ change/modify/ alter this filing and make further submissions as may be required at later stage as the filing is being done based on the best available information.
- o) Treat the filing as complete in view of substantial compliance as also the specific requests for waivers with justification placed on record.

BY THE APPLICANT THROUGH

**CHIEF ENGINEER,
PLANNING CUM REGULATORY
HPGCL, PANCHKULA**